

Mod-5 Galois images from genus 2 Jacobians

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Abstract

For J a Jacobian of a genus 2 curve, the Galois representation $\rho_{J,\ell} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{Aut}(J[\ell]) \simeq \text{GSp}_4(\mathbb{F}_\ell)$ is typically surjective, with smaller images indicating extra arithmetic structure. It is already known how to probabilistically compute whether $\rho_{J,\ell}$ is surjective, and recent work by Chidambaram computes $\text{im } \rho_{J,\ell}$ for $\ell = 2, 3$. We use endomorphism-agnostic methods to probabilistically compute $\rho_{J,5}$ for the Jacobians of 98% of genus 2 curves in the L-functions and Modular Forms Database (LMFDB) for which $\rho_{J,5}$ is not yet known. For the remaining Jacobians, we give a short list of locally conjugate candidate images.

Introduction

Throughout the paper, let all curves be *nice* in the sense of Poonen, i.e., smooth, projective, and geometrically integral. Let $C/\mathbb{Q}$ be a nice genus g curve. Let $J = \text{Jac}(C)$ be the Jacobian of C . There is a natural action of the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on C , which extends to an action on J .

Let $J[n]$ denote the n -torsion subgroup of J . The action of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on J restricts to an action on $J[n]$, and this action preserves the group law and Weil pairing; this gives a map $\rho_{J,n} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{Aut}(J[n])$. For $n = \ell$ a prime, we identify $\text{Aut}(J[n]) \cong \text{GSp}_{2g}(\mathbb{F}_\ell)$.

Definition 0.0.1 (Mod- ℓ Galois representation from torsion). For a Jacobian J of a genus g curve and a prime ℓ , the *Galois representation from ℓ -torsion on J* is the map $\rho_{J,\ell} : \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GSp}_{2g}(\mathbb{F}_\ell)$.

The image of the mod- ℓ Galois representation is of interest because it encodes various arithmetic information. For example, for J the Jacobian of a genus 2 curve, J admits a rational ℓ -torsion point if and only if $\text{im } \rho_{J,\ell}$ is conjugate to a subgroup of G_1 , and J admits a rational ℓ -isogeny if and only if $\text{im } \rho_{J,\ell}$ is conjugate to a subgroup of G_2 , where

$$G_1 = \begin{bmatrix} 1 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix} \quad \text{and} \quad G_2 = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix}.$$

It is difficult to rigorously determine the image of Galois, but there is a large body of work using probabilistic methods to determine high-likelihood candidate images. Sutherland [1] computed $\text{im } \rho_{E,\ell}$ using probabilistic methods for all elliptic curves E without complex multiplication in the LMFDB and arbitrary primes ℓ . For genus 2 curves, it is a theorem of Serre [2] that if $\text{End } J(\mathbb{Q}) = \mathbb{Z}$ then $\rho_{J,\ell}$ surjects onto $\text{GSp}_4(\mathbb{F}_\ell)$ for all but finitely many primes. For each such curve in the LMFDB, Banwait, Brumer, Kim, Klagsbrun, Mayle, Srinivasan, and Vogt [3] developed a probabilistic algorithm to compute the finite set of non-surjective primes. Some work towards computing the precise image in non-surjective cases already exists. In the $\ell = 2$ case, it suffices to consider the Weierstrass points to identify $\mathbb{Q}(J[2])$. The $\ell = 3$ case is addressed in recent work by Chidambaram [4].

In this paper, we address the $\ell = 5$ case. Our main contribution lies in developing an algorithm to compute the likely image of $\rho_{J,5}$ for a Jacobian J .

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Theorem (Main Theorem). Let $C/\mathbb{Q}$ be a genus 2 curve with Jacobian J . Fix any $a \in \mathbb{Z}$. There exists an effective constant N , depending only on C , such that there exists an algorithm sampling all primes in the range $[a, N]$ that produces a list containing the mod-5 image of Galois. The list is composed of at most eight subgroups of $\mathrm{GSp}_4(\mathbb{F}_5)$, all of equal order.

The parameter N in the Main Theorem is a computationally intractable bound. Hence, we compute likely images by sampling a smaller set of primes.

Computational Results. Let J be the Jacobian of one of the 3990 genus 2 curves in the [5] for which $\rho_{J,5}$ is not known to be surjective. On input J , Algorithm 1 produces a set of at most five equal-order likely images of $\rho_{J,5}$. Furthermore, for 3920 of the curves described above, Algorithm 1 produces a single likely image.

Remark 0.0.2. The algorithm does not always determine the image up to isomorphism; the two candidate images for the curve with LMFDB label 37500.a.37500.1 are non-isomorphic. Furthermore, the algorithm does not determine the image up to local conjugacy, as the three candidate images for the curve with LMFDB label 69696.c.627264.1 are pairwise non-locally conjugate. See Sections 4.2 and 4.6, as well as Remark 3.1.3 for further discussion.

Remark 0.0.3. Our method is agnostic to extra endomorphisms and splitting behavior of Jacobians. Leveraging such information would require very different techniques, and would likely resolve some, but definitely not all, of the still undetermined images (see Sections 4.2, 4.7).

Theorem (Likelihood of error). Assume an independent random distribution of Frobenius elements and that, as a prior, every subgroup of $\mathrm{GSp}_4(\mathbb{F}_5)$ with surjective similitude character is an equally likely candidate image. Then, the probability that $\rho_{J,5}$ is *not* on the list of subgroups output by Algorithm 1 is bounded above by 1 in 9.5 billion. If we instead set prior probabilities in accordance with the proportion of LMFDB curves computed to have each image, then the probability of error is bounded above by 6.54 billion.

In Section 1 we review key background about the structure of Jacobians of genus 2 curves and their 5-torsion subgroups. Namely, we describe the possible images of Galois, how to explicitly represent points on a Jacobian using Mumford coordinates, and a bound on the height of torsion. We also review Frobenius elements and Chebotarev density in the context of Galois action on torsion. In Section 2, we prove a lemma which ensures correctness of Algorithm 5. In Section 3, we describe Algorithm 1 in detail and provide a proof of the Main Theorem. In Section 4, we prove the error bound claimed in the Computational Results Theorem, present the result of running Algorithm 1 on the genus 2 curves in the LMFDB, and give some additional information about those images.

Code for this work can be found in our GitHub repository

<https://GitHub.com/maathilde-k/Mod-5-Galois-Images-of-Genus-2-Abelian-Curves/>.

Throughout the paper, let $f \in \mathbb{Z}[x]$ be a squarefree polynomial of degree 5 or 6, C be the corresponding genus 2 hyperelliptic curve given by $y^2 = f(x)$, and $J = \mathrm{Jac}(C)$ be the Jacobian of C . Let $\mathrm{im} \rho_{J,5}$ denote the image of the associated mod-5 Galois representation, and refer to this as the “image of Galois”. Generally, one has the $\overline{\mathbb{Q}}$ -endomorphism group $\mathrm{End}(J) \cong \mathbb{Z}$ with the endomorphisms being multiplication-by- n maps. We refer to curves C with $\mathrm{End}(\mathrm{Jac}(C_{\overline{\mathbb{Q}}})) = \mathbb{Z}$ as *typical*. We refer to curves whose Jacobians admit additional endomorphisms as *atypical*.

Throughout our paper, we label subgroups of $\mathrm{GSp}_4(\mathbb{F}_\ell)$ according to the labeling scheme used by [5]. These labels take the format $\ell.i.j$, where ℓ is the characteristic of the ground field $\mathbb{F}_\ell$. The second parameter, i , indicates the index of the subgroup in $\mathrm{GSp}_4(\mathbb{F}_\ell)$. The final parameter, j , is simply a position in the list of conjugacy classes of index i subgroups.

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1. Background

1.1. The possible images of Galois

The image of $\rho_{J,5}$ is a subgroup of $\mathrm{GSp}_4(\mathbb{F}_5)$. Over $\mathbb{Q}$, the image of the Galois representation must have surjective similitude character, restricting the possibilities for $\mathrm{im} \rho_{J,5}$. Additionally, complex conjugation acts on $J[5]$ as a non-trivial involution and has similitude -1 (since conjugation sends $\zeta_5 \mapsto \zeta_5^{-1}$). Thus, the image of Galois must include an element of order 2 and similitude -1 . This leaves (up to conjugacy) 1125 subgroups of $\mathrm{GSp}_4(\mathbb{F}_5)$ for consideration. We therefore work on distinguishing between these 1125 subgroups.

1.2. Mumford Coordinates

Throughout the paper, we denote points on the Jacobian by their *Mumford coordinates*.

Definition 1.2.1 (Mumford coordinates). Let C be a genus 2 curve given by $y^2 = f(x)$. The curve C contains two (possibly coincident) points ∞_1 and ∞_2 at infinity. Let D be a divisor class on C , with reduced divisor $[P_1] + [P_2] - ([\infty_1] + [\infty_2])$, where P_1, P_2 are points on C . The *Mumford coordinates* (u, v) of D are the unique pair of polynomials u and v such that:

1. u is monic of degree at most 2,
2. $u(x(P_1)) = 0$ and $u(x(P_2)) = 0$,
3. v is of degree at most 1,
4. $u \mid v^2 - f$.

Remark 1.2.2. If P_1 or P_2 lies at infinity, u has degree less than 2. If $P_1 = P_2$, u has a double root at their x coordinate.

Remark 1.2.3. A point on the Jacobian is defined over K if and only if all the coefficients of the polynomials comprising its Mumford coordinates lie in K .

Remark 1.2.4. Mumford coordinates can be defined for Jacobians of hyperelliptic curves of any genus.

1.3. Frobenius Elements

The most direct approach to computing the image of a Galois representation would be to compute the ℓ -torsion points over $\overline{\mathbb{Q}}$ and their images under different field automorphisms. However, the 5-torsion field of an abelian surface is too large to compute explicitly for a large number of curves. Taking inspiration from [1] and [3], we instead collect information by sampling Frobenius elements.

Definition 1.3.1 (5-torsion field). Let C be a genus 2 curve defined over $\mathbb{Q}$ with Jacobian $J = \mathrm{Jac}(C)$. The *5-torsion field of J* , denoted $\mathbb{Q}(J[5])$, is the smallest field over which all the 5-torsion points are defined.

Concretely, the 5-torsion field can be obtained by adjoining to $\mathbb{Q}$ the coefficients of the Mumford coordinates of the 5-torsion points.

Remark 1.3.2. The 5-torsion field is a Galois extension.

Let p be a prime of good reduction. The action of the Frobenius element Frob_p on the 5-torsion of an abelian variety J is the same as the action of the Frobenius automorphism of $\overline{\mathbb{F}}_p$ on the 5-torsion $J(\overline{\mathbb{F}}_p)[5]$ over $\mathbb{F}_p$. Thus, the properties of Frobenius elements can be calculated indirectly from the structure of the reduction of J modulo p . We refer to such calculations as “local methods”. Frobenius elements are representative of the Galois group in the sense that their images under $\rho_{J,5}$ equidistribute in conjugacy classes, as formalized in the following theorem.

Theorem 1.3.3 (Chebotarev Density Theorem). Let $K/\mathbb{Q}$ be a Galois extension. Let $\mathcal{C}$ be a conjugacy class of $G = \text{Gal}(K/\mathbb{Q})$. Let

$$P_{\mathcal{C}}(n) = \#\{\text{prime } p \mid \text{Frob}_p \in \mathcal{C}, p < n\}$$

denote the number of primes p less than n such that $\text{Frob}_p \in \mathcal{C}$. Let $\pi(n)$ denote the number of primes less than n . Then, we have

$$\lim_{n \rightarrow \infty} \frac{P_{\mathcal{C}}(n)}{\pi(n)} = \frac{\#\mathcal{C}}{\#G}.$$

By applying the Chebotarev density theorem to $K = \mathbb{Q}(J[5])$, we may use statistical methods to infer the likely image of the Galois representation. Effective versions of the Chebotarev density theorem, such as that in [6], give bounds on the error of such statistical methods. Thus, adequately large samples could give results not just with arbitrarily small probability of error, but with 0 probability of error. However, even assuming the generalized Riemann Hypothesis, the required sample sizes are intractably large.

We compute the characteristic polynomial of the Frobenius element Frob_p through the L -polynomial of the reduction of $J \bmod p$. By the Weil Conjectures, the L -polynomial of $J(\mathbb{F}_p)$ is the reciprocal polynomial of the characteristic polynomial of the Frobenius element Frob_p . L -polynomials can be efficiently computed using existing software such as Kedlaya and Sutherland's [7] `Smalljac` program.

Another piece of local information we extract from Frob_p is the dimension of its 1-eigenspace as an automorphism of $J[5] \cong \mathbb{F}_5^4$. Because only the points defined over $\mathbb{F}_p$ are fixed by the Frobenius automorphism, one may compute the dimension of the 1-eigenspace of Frob_p by counting the number of 5-torsion points over $\mathbb{F}_p$.

Remark 1.3.4. Implicitly, the distribution of these dimensions over many sampled primes (the “1-eigenspace spectrum”), encapsulates information about the overall order of the group. A matrix is the identity if and only if it has a 4-dimensional 1-eigenspace. Thus, one may estimate the order of the group $\text{im } \rho_{J,5}$ by taking the reciprocal of the proportion of sampled Frobenius elements with a 4-dimensional 1-eigenspace.

Lemma 1.3.5 (To be used in proof of 3.1.10). Among all elements of $\text{GSp}_4(\mathbb{F}_5)$, there are exactly 99 distinct (characteristic polynomial, 1-eigenspace dimension) pairs.

Proof. This is a straightforward computation; see the [Paper Lemmas](#) folder in the GitHub repository. $\square$

1.4. Height Bounds for Torsion Points

A main technique of this paper is the computation of 5-torsion subgroups over quadratic fields via $\mathfrak{p}$ -adic lifting and rational recognition. Heights on Jacobians and bounds on the height of torsion are important tools in constraining the points over which to search. This section reviews the relevant definitions and cites a height bound theorem used in our search.

Definition 1.4.1 ((Logarithmic) heights in $\mathbb{P}^n(K)$). Let K be a number field, and let $P = [x_0 : \dots : x_n] \in \mathbb{P}^n(K)$. Let M_K denote the set of places of K , and let $\{|\cdot|_v : v \in M_K\}$ denote a set of absolute values on K , normalized so that a product formula holds. Then, the (logarithmic) height $h_{\mathbb{P}^n(K)}(P)$ of P is given by

$$h_{\mathbb{P}^n(K)}(P) = \frac{1}{[K : \mathbb{Q}]} \log \left(\prod_{v \in M_K} \max\{|x_0|_v, \dots, |x_n|_v\} \right).$$

Remark 1.4.2. The height $h_{\mathbb{P}^n(K)}(P)$ as defined above is independent of choice of representative of P precisely because the absolute values are normalized towards a product formula. The quotient by $[K : \mathbb{Q}]$ in $h_{\mathbb{P}^n(K)}$ is not essential in our context, but is nice in that it extends to a height on $\mathbb{P}^n(\overline{\mathbb{Q}})$.

Definition 1.4.3 (Kummer surface associated to a Jacobian). The Jacobian admits a negation involution $P \mapsto -P$. The quotient of J by this involution is ramified precisely at the 2-torsion. The image of the quotient map is the *associated Kummer surface*.

Definition 1.4.4 (Standard embedding of the Kummer surface). Let J be the Jacobian of the genus 2 curve $C : y^2 = f(x)$, with the x^n coefficient of f denoted by a_n . Let $\phi : J \rightarrow \mathbb{P}^3$ be the unique map which, on the dense open set where the following makes sense, is given by

$$\phi : P = (x^2 + \alpha x + \beta, \delta x + \gamma) \mapsto [1 : -\alpha : \beta : m],$$

where

$$m = \frac{2a_6\beta^3 - a_5\alpha\beta^2 + 2a_4\beta^2 - a_3\alpha\beta + a_2\beta - a_1\alpha + 2a_0 - 2(\gamma^2\beta - \gamma\delta\alpha + \delta^2)}{\alpha^2 - 4\beta}.$$

This map, first appearing in [8, pp. 6–19], is the *standard embedding* of the Kummer Surface associated to J .

Definition 1.4.5 (Naïve height). Let J be the Jacobian of a genus 2 hyperelliptic curve over $\mathbb{Q}$, and let ϕ be the standard embedding of its associated Kummer surface. Then, the naïve height $h(P)$ of a point $P \in J(K)$ is given by $h(P) = h_{\mathbb{P}^3(K)}(\phi(P))$.

Remark 1.4.6. Note that there are other possible embeddings of the Kummer surface associated to J into $\mathbb{P}^3$, and the naïve height is dependent, up to a constant factor, on the particular embedding chosen. The explicit embedding in Definition 1.4.4 is common in the literature; see [9, pp. 184–185][10, p. 334].

Theorem 1.4.7 (Corollary 8.1 in [9]). Let K be a number field. Let J be the Jacobian of the genus 2 curve C/K given by $C : y^2 = f(x)$. Fix $d \in \mathbb{Z}_+$. There is an effectively computable upper bound B , dependent only on f and d , on the naïve height of any torsion point of J defined over a degree d number field.

The cited paper includes an explicit formula for this bound.

2. Height Lemmas

The following lemmas, culminating in Theorem 2.0.6, are used to guarantee correctness of Algorithm 5. We recommend skipping this section on a first reading.

Lemma 2.0.1. Let $I \subseteq \mathcal{O}_K$ be a prime of a number field. Let $S(n, B)$ be the set of elements of $\mathbb{P}^n(K)$ with (multiplicative) height bounded by B . Then, the base change map $\mathbb{P}^n(K) \rightarrow \mathbb{P}^n(\mathcal{O}_K/I)$ is injective on $S(n, B)$ if and only if it is injective on $S(1, B)$.

Proof. If the base change is injective on $S(n, B)$, then it is injective on $S(1, B)$ by restriction. Conversely, suppose reduction maps distinct points $P_1 = [x_0 : \dots : x_n]$ and $P_2 = [y_0 : \dots : y_n]$ to the same point of $\mathbb{P}^n(\mathcal{O}_K/I)$. Let k be an index where $x_k, y_k \neq 0$. Since $P_1 \neq P_2$, there is some index pair (i, j) for which $x_i y_j \neq x_j y_i$. It cannot be that both $x_i y_k = x_k y_i$ and $x_j y_k = x_k y_j$, so without loss of generality conclude $x_i y_k \neq x_k y_i$. Since $x_k, y_k \neq 0$, we may consider the points $[x_i : x_k], [y_i : y_k] \in \mathbb{P}_1(K)$. It is easy to see that dropping all but two coordinates weakly decreases height, so $[x_i : x_k], [y_i : y_k] \in S(1, B)$. Furthermore, $[x_i : x_k]$ and $[y_i : y_k]$ reduce to the same point in $\mathbb{P}_1(\mathcal{O}_K/I)$. $\square$

Lemma 2.0.2. Let K be a degree d number field with discriminant Δ_K , satisfying $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$. Let $M_K = \frac{d!}{d^d} \left(\frac{4}{\pi}\right)^{r_2} \sqrt{|\Delta_K|}$ denote the Minkowski bound for ideal classes of K . $I \subseteq K$ be a nonzero integral ideal with norm $N(I) = |\mathcal{O}_K/I|$. Let

$$B = \frac{\sqrt{N(I)}}{2^{d/2} M_K}.$$

Then, the reduction map $\mathbb{P}^n(K) \rightarrow \mathbb{P}^n(\mathcal{O}_K/I)$ is injective on $S(n, B)$ for any positive integer n .

Proof. By Lemma 2.0.1, we may assume $n = 1$. Suppose P, Q are distinct points of $\mathbb{P}_1(K)$ with the same reduction in $\mathbb{P}_1(\mathcal{O}_K/I)$. We may take “nearly primitive” integral representatives $P = [x_0 : x_1], Q = [y_0 : y_1]$, meaning all coordinates lie in $\mathcal{O}_K$ and $N(x_0 \mathcal{O}_K + x_1 \mathcal{O}_K), N(y_0 \mathcal{O}_K + y_1 \mathcal{O}_K) \leq M_K$. Distinctness and same reduction imply that $\alpha = x_0 y_1 - x_1 y_0 \in I \setminus \{0\}$. Note that $N(\alpha) \geq N(I)$, since $I \mid (\alpha)$.

We now upper-bound $N(\alpha)$ in terms of $\max\{H(P), H(Q)\}$. Since we are using nearly primitive representatives, we have control over the non-archimedean contributions to height. Specifically,

$$\begin{aligned}
N(\alpha) &= \prod_{\sigma: K \rightarrow \mathbb{C}} \sigma(\alpha) = \prod_{\sigma: K \rightarrow \mathbb{C}} |\sigma(x_0 y_1 - x_1 y_0)| \\
&= \prod_{\sigma: K \rightarrow \mathbb{C}} |\sigma(x_0)\sigma(y_1) - \sigma(x_1)\sigma(y_0)| \\
&\leq \prod_{\sigma: K \rightarrow \mathbb{C}} 2 \cdot \max\{|\sigma(x_0)|, |\sigma(x_1)|\} \cdot \max\{|\sigma(y_0)|, |\sigma(y_1)|\} \\
&= 2^d \prod_{v|\infty} \max\{|x_0|_v, |x_1|_v\} \prod_{v|\infty} \max\{|y_0|_v, |y_1|_v\} \\
&\leq 2^d M_K H(P) M_K H(Q) = 2^d M_K^2 H(P) H(Q) \leq 2^d M_K^2 \max\{H(P), H(Q)\}^2.
\end{aligned}$$

Stringing things together and solving for $\max\{H(P), H(Q)\}$, we conclude

$$\max\{H(P), H(Q)\} \geq \frac{\sqrt{N(I)}}{2^{d/2} M_K},$$

proving the theorem. □

Lemma 2.0.3. Let $d = [K : \mathbb{Q}]$. Let $P \in \mathbb{P}^3(K)$ have multiplicative height $H(P)$. Among all possible choices of integral homogeneous coordinates $(a_0, a_1, a_2, a_3) \in \mathcal{O}_K^4$ for P , let $\vec{P}$ denote the representative whose embedding in fourth-power Minkowski space has minimal L_2 norm. Then,

$$\|\vec{P}\|_2 \leq 2\sqrt{d} \left((\sqrt{|\Delta_K|} H(P))^{1/d} \right).$$

Proof. We will proceed by showing that the set of interest forms a rank d lattice, compute the covolume, and then use Minkowski's theorem. Let (a_0, a_1, a_2, a_3) be any choice of integral homogeneous coordinates for P . Let $\mathfrak{c} = \langle a_0, a_1, a_2, a_3 \rangle$ be their content. The possible integral coordinates for P are exactly the set $\{(ca_0, ca_1, ca_2, ca_3) : c \in \mathfrak{c}^{-1}\}$. Let $\sigma : K \rightarrow K \otimes \mathbb{R} \cong \mathbb{R}^d$ denote the canonical embedding into Minkowski space, and let $e_P : \mathbb{R}^d \rightarrow \mathbb{R}^{4d}$ be given by $v \mapsto (v \odot \sigma(a_0), v \odot \sigma(a_1), v \odot \sigma(a_2), v \odot \sigma(a_3))$, where $\odot$ denotes the Hadamard product of vectors in $\mathbb{R}^d$. The lattice of embeddings of integral choices of coordinates for P is the image $L := (e_P \circ \sigma)(\mathfrak{c}^{-1})$. The covolume of this lattice (in its d -dimensional span) may be computed as $\left(\sqrt{|\Delta_K|} / N(\mathfrak{c}) \right) \sqrt{\det([e_P]^\top [e_P])}$, where $[e_P]$ is the matrix for e_P in the standard basis. Let $\sigma_1, \dots, \sigma_{r_1}$ be the real embeddings of K , and $\tau_1, \dots, \tau_{r_2}$ be non-complex-conjugate complex embeddings of K . Unpacking, we can express $[e_P]$ as the block matrix

$$[e_P] = \begin{bmatrix} A_0 \\ A_1 \\ A_2 \\ A_3 \end{bmatrix}$$

where each A_i is a diagonal matrix given by

$$A_i = \begin{bmatrix} \sigma_1(a_i) & & & & & \\ & \dots & & & & \\ & & \sigma_{r_1}(a_i) & & & \\ & & & \Re(\tau_1(a_i)) & & \\ & & & & \Im(\tau_1(a_i)) & \\ & & & & & \dots \\ & & & & & & \Re(\tau_{r_2}(a_i)) \\ & & & & & & & \Im(\tau_{r_2}(a_i)) \end{bmatrix}.$$

The determinant of $[e_P]^\top [e_P]$ can be easily bounded as

$$\begin{aligned}
\det([e_P]^\top [e_P]) &= \prod_{\sigma} \sqrt{\sigma(a_0)^2 + \sigma(a_1)^2 + \sigma(a_2)^2 + \sigma(a_3)^2} \\
&\quad \cdot \prod_{\tau} \sqrt{|\Re(\tau(a_0))|^2 + |\Re(\tau(a_1))|^2 + |\Re(\tau(a_2))|^2 + |\Re(\tau(a_3))|^2} \\
&\quad \cdot \prod_{\tau} \sqrt{|\Im(\tau(a_0))|^2 + |\Im(\tau(a_1))|^2 + |\Im(\tau(a_2))|^2 + |\Im(\tau(a_3))|^2} \\
&\leq \prod_{\sigma} \sqrt{\sigma(a_0)^2 + \sigma(a_1)^2 + \sigma(a_2)^2 + \sigma(a_3)^2} \\
&\quad \cdot \prod_{\tau} (|\tau(a_0)|^2 + |\tau(a_1)|^2 + |\tau(a_2)|^2 + |\tau(a_3)|^2) \\
&\leq \prod_{\sigma} \sqrt{4 \max\{|\sigma(a_i)|\}} \cdot \prod_{\tau} 4 \max\{|\tau(a_i)|\} \\
&\leq 2^d \prod_{v|\infty} \max\{|a_i|_v\} = 2^d N(\mathfrak{c}) H(P).
\end{aligned}$$

Thus, we obtain the bound

$$\text{covol}(L) = \left(\sqrt{|\Delta_K|/N(\mathfrak{c})} \right) \sqrt{\det([e_P]^\top [e_P])} \leq \left(\sqrt{|\Delta_K|/N(\mathfrak{c})} \right) (2^d N(\mathfrak{c}) H(P)) = 2^d \sqrt{|\Delta_K|} H(P).$$

The Minkowski bound then yields

$$\|\vec{P}\|_2 \leq \sqrt{d} \left(2^d \sqrt{|\Delta_K|} H(P) \right)^{1/d} = 2\sqrt{d} \left(\sqrt{|\Delta_K|} H(P) \right)^{1/d},$$

as desired. □

Lemma 2.0.4. Let $[a_0 : a_1 : a_2 : a_3]$ be integral homogeneous coordinates for $Q \in \mathbb{P}^3(K)$. Then,

$$\|(a_0, a_1, a_2, a_3)\|_2 \geq \sqrt{\frac{d}{2}} H(Q)^{1/d}.$$

Proof. We simply compute

$$\begin{aligned}
\|(a_0, a_1, a_2, a_3)\|_2 &= \sqrt{\sum_{\sigma: K \rightarrow \mathbb{C}} \frac{|\sigma(a_0)|^2 + |\sigma(a_1)|^2 + |\sigma(a_2)|^2 + |\sigma(a_3)|^2}{\dim_{\mathbb{R}} \text{Span}_{\mathbb{R}}(\sigma(K))}} \\
&\geq \sqrt{\frac{1}{2} \sum_{\sigma: K \rightarrow \mathbb{C}} (|\sigma(a_0)|^2 + |\sigma(a_1)|^2 + |\sigma(a_2)|^2 + |\sigma(a_3)|^2)} \\
&\geq \sqrt{\frac{1}{2} \sum_{\sigma: K \rightarrow \mathbb{C}} \max_i \{|\sigma(a_i)|\}^2} \\
&\geq \sqrt{\frac{d}{2}} \left(\prod_{\sigma: K \rightarrow \mathbb{C}} \max_i \{|\sigma(a_i)|\} \right)^{1/d} \\
&= \sqrt{\frac{d}{2}} \left(\prod_{v|\infty} \max_i \{|a_i|_v\} \right)^{1/d} \\
&\geq \sqrt{\frac{d}{2}} H(P)^{1/d},
\end{aligned}$$

where to go from the third line to the fourth uses AM-GM, and the last inequality uses the hypothesis that $a_i \in \mathcal{O}_K$. □

Lemma 2.0.5. Let I be an ideal of $\mathcal{O}_K$ with norm $N(I)$. Let $\vec{0} \neq (a_0, a_1, a_2, a_3) \in I \times I \times I \times I$. Then, $\|(a_0, a_1, a_2, a_3)\|_2 \geq \sqrt{\frac{d}{2}} N(I)^{1/d}$.

Proof. This proof is similar to that of Lemma 2.0.4. Let $a := a_0$ and compute that

$$\begin{aligned} \|(a_0, a_1, a_2, a_3)\|_2 &\geq \|a\|_2 = \sqrt{\sum_{\sigma: K \rightarrow \mathbb{C}} \frac{|\sigma(a)|^2}{\dim_{\mathbb{R}} \text{Span}_{\mathbb{R}}(\sigma(K))}} \\ &\geq \sqrt{\frac{1}{2} \sum_{\sigma: K \rightarrow \mathbb{C}} |\sigma(a)|^2} \\ &\geq \sqrt{\frac{d}{2}} \left(\prod_{\sigma: K \rightarrow \mathbb{C}} |\sigma(a)| \right)^{1/d} \\ &= \sqrt{\frac{d}{2}} |N(a)|^{1/d} \geq \sqrt{\frac{d}{2}} N(I)^{1/d}. \end{aligned}$$

□

Theorem 2.0.6. Let $P \in \mathbb{P}^3(K)$ be a point with multiplicative height bounded by C . Let $q = N(\mathfrak{p})$. Let

$$r = \left\lceil \frac{2}{\log q} \left((2d^2 + \frac{3}{2}d) \log 2 + \frac{1}{2} \log |\Delta_K| + \log M_K + \log C \right) \right\rceil.$$

Let (a_0, a_1, a_2, a_3) be integral projective coordinates for a point $Q \in \mathbb{P}^3(K)$ which agrees with $P \bmod \mathfrak{p}^r$. Let $\{\alpha_1, \dots, \alpha_f\}$ be representatives in $\mathcal{O}_K$ for an $\mathbb{F}_p$ basis of the residue field $\mathcal{O}_K/\mathfrak{p}$, where p is the characteristic. Let $\{b_1, \dots, b_d\}$ be a $\mathbb{Z}$ -basis for $\mathfrak{p}^r$. Take the lattice L generated by $\{(\alpha_i a_0, \alpha_i a_1, \alpha_i a_2, \alpha_i a_3)\}$ together with each $(b_{i_1}, b_{i_2}, b_{i_3}, b_{i_4})$ inside the fourth power of the Minkowski space of K . Then, applying LLL to L with $\delta = 0.75$ will recover a representation of P as a shortest basis vector.

Proof. Observe that the generated lattice consists exactly of those points in $\mathcal{O}_K^4$ which either

1. are the homogeneous coordinates of a point in $\mathbb{P}^3(K)$ whose reduction mod $\mathfrak{p}^r$ agrees with that of P or
2. are contained in $\mathfrak{p}^r \times \mathfrak{p}^r \times \mathfrak{p}^r \times \mathfrak{p}^r$.

Call non-zero lattice points “type-1” or “type-2” depending on which of the above cases they fall under. By Lemma 2.0.3, there is a lattice point representing P with norm at most $2\sqrt{d} \left(\sqrt{|\Delta_K|} C \right)^{1/d}$, so LLL will return a lattice point with norm at most $2^{(4d-1)/2} \cdot 2\sqrt{d} \left(\sqrt{|\Delta_K|} C \right)^{1/d}$. By Lemma 2.0.2, all other points of $\mathbb{P}^3(K)$ agreeing with $P \bmod \mathfrak{p}^r$ have multiplicative height at least

$$H(Q) \geq \frac{q^{r/2}}{2^{d/2} M_K} \geq 2^{2d^2+d} \sqrt{|\Delta_K|} C.$$

So, by Lemma 2.0.4, any type-1 lattice point representing a point $Q \neq P$ has norm at least

$$\|v\|_2 \geq \sqrt{\frac{d}{2}} H(Q)^{1/d} \geq \sqrt{\frac{d}{2}} \left(2^{2d+1} \left(\sqrt{|\Delta_K|} C \right)^{1/d} \right),$$

and thus will not be returned.

Similarly, by Lemma 2.0.5, type-2 points have norm at least

$$\|v\|_2 \geq \sqrt{\frac{d}{2}} q^{r/d} \geq 2^{4d+\frac{5}{2}} \sqrt{d} \left(\sqrt{|\Delta_K|} C \right)^{2/d},$$

which is far too large to be the output of LLL.

□

3. The Algorithm

Given a genus 2 curve C , we compute two types of information about C — local information and global information. Local information is data obtained from sampling Frobenius elements, namely their characteristic polynomials and 1-eigenspace dimensions. Global information consists of data obtained by examining $J = \text{Jac}(C)$ over number fields. These data are used to narrow down the possible images of $\rho_{J,5}$, as laid out in Algorithm 1 below. The algorithm is implemented in Magma [11] and is available at the following repository:

<https://GitHub.com/maathilde-k/Mod-5-Galois-Images-of-Genus-2-Abelian-Curves/tree/main>.

To illustrate the algorithms described, we employ the curve $y^2 = 4x^5 - 20x^3 + 5x^2 + 20x - 4$ (LMFDB label 431250.a.431250.1) as a running example.

Algorithm 1: An algorithm to compute a short list of likely candidates for the mod-5 image of Galois.

Input: The Jacobian J of a genus 2 curve $C : y^2 = f(x)$, a range $[a, b]$, and a confidence threshold ν .

Output: A short list of subgroups of $\text{GSp}_4(\mathbb{F}_5)$, which likely (and for large enough b, ν , provably) contains the image of the mod-5 Galois representation associated to the Jacobian of C .

1. Compute the subgroup lattice of $\text{GSp}_4(\mathbb{F}_5)$ and initialize **possibilities** as the set of all (conjugacy classes of) subgroups with surjective similitude character and order 2 element with similitude character -1 (corresponding to complex conjugation).
2. Set $U = 2a$.
3. **Do**
 - (a) Using Algorithm 2, compute the distribution of (characteristic polynomial, 1-eigenspace dimension) pairs among Frobenius elements associated to each good prime in $[a, U]$.
 - i. For each sampled (characteristic polynomial, 1-eigenspace dimension) pair, keep only the subgroups in **possibilities** containing an element realizing this invariant.
 - (b) Using Algorithm 3, compute the set of most likely possible images and a confidence parameter η (see Section 3.1.2, particularly Definition 3.1.13 for details).
 - (c) Increase U by a .

Until $\eta > \nu$ or $U \geq b$.

4. Compute $n_1(J) := \#J(\mathbb{Q})[5]$.
 5. Using Algorithm 8, compute the number $n_2(J)$ of quadratic 5-torsion points.
 6. Remove from **possibilities** any subgroup which does not act on $\mathbb{F}_5^4$ with n_1 orbits of size 1 and $n_2/2$ orbits of size 2.
 7. **Return** **possibilities**.
-

3.1. Local Information

Definition 3.1.1 (Local distribution (of a group)). Given a finite matrix group G over a field F , the *local distribution* of G is a probability mass function $\mathcal{F}_G : F[x] \times \mathbb{Z} \rightarrow \mathbb{R}$ where

$$\mathcal{F}_G(f, n) = \frac{1}{\#G} \cdot \# \{M \in G : \text{CharacteristicPolynomial}(M) = f \text{ and } 1\text{-eigenspace-dimension}(M) = n\}.$$

That is, the local distribution assigns to each (characteristic polynomial, 1-eigenspace dimension) pair the probability of obtaining that pair when sampling elements uniformly from the group G .

Definition 3.1.2 (Local conjugacy). We say two subgroups $H_1, H_2 \leq G$ are *locally conjugate* if $|H_1 \cap C| = |H_2 \cap C|$ for all conjugacy classes C of G .

Remark 3.1.3. In $\mathrm{GL}_2(\mathbb{F}_\ell)$, the local distribution of a group determines the group up to isomorphism and up to local conjugacy [1, Corollary 3.30]. In $\mathrm{GSp}_4(\mathbb{F}_\ell)$, neither of these determinations hold. Thus, there is not much more that can be said relating candidate images beyond Remark 1.3.4.

Lemma 3.1.4. Up to conjugacy, no more than eight subgroups of $\mathrm{GSp}_4(\mathbb{F}_5)$ with surjective similitude character all share the same local distribution.

Proof. The code used to compute this fact can be found in the [Paper Lemmas](#) folder in the GitHub repository. $\square$

Remark 3.1.5. We will refer to the local distribution $\mathcal{F}_{\mathrm{im} \rho_{J,5}}$ of the image of Galois associated to J as the *true local distribution* of J . This is in contrast to the below defined empirical distribution.

Definition 3.1.6 (Empirical local distribution). Given a Jacobian J and a range $[a, b]$ containing at least one prime, the *empirical local distribution* $\mathcal{E}(J, a, b)$ is the distribution of the pair (characteristic polynomial, 1-eigenspace dimension) associated to Frob_p acting on $J[5]$, when p is drawn from primes between a and b inclusive for which J has good reduction at p .

Remark 3.1.7. Note that characteristic polynomials and dimensions of 1-eigenspaces are conjugation invariant, so it is not an issue that Frobenius elements are only defined up to conjugacy.

Remark 3.1.8. By the Chebotarev Density Theorem, for any fixed a the empirical local distribution $\mathcal{E}(J, a, b)$ converges to the true local distribution $\mathcal{F}_{\mathrm{im} \rho_{J,5}}$ as $b \rightarrow \infty$. Thus, for large enough N , the true distribution is that which is nearest the empirical distribution. This is made precise in Lemma 3.1.10.

We make use of this in two steps. Algorithm 2 computes the empirical local distribution. Then, Algorithm 3 selects the subgroups with local distributions best resembling the empirical local distribution computed from Algorithm 2.

3.1.1. Sampling Algorithm

Algorithm 2 computes the empirical local distribution by sampling Frobenius elements corresponding to primes in the interval $[a, b]$. We compute the characteristic polynomials using [7] and the dimensions of 1-eigenspaces by counting 5-torsion points of the Jacobian over $\mathbb{F}_p$. The algorithm is formalized below.

Algorithm 2: An algorithm to compute the empirical local distribution by sampling Frobenius elements

Input: A Jacobian J of a genus 2 curve and a range $[a, b]$ over which to sample primes.

Output: A dictionary `dist` whose keys are (characteristic polynomial, 1-eigenspace dimension) tuples and whose values are the corresponding empirical frequencies.

1. Initialize the multiset `invariant_counts`.
 2. **For** all primes $p \in [a, b]$ of good reduction (i.e. not dividing the conductor of J), add the tuple $(\Pi_p, \dim_p)$ to `invariant_counts`, where Π_p and $\dim_p$ are respectively the characteristic polynomial and the dimension of the 1-eigenspace of Frob_p .
 3. Initialize the dictionary `dist`.
 4. **For** all pairs $(\Pi, \dim)$ in `invariant_counts` **do**
 - (a) Add to `dist` a key-value pair whose key is $(\Pi, \dim)$ and whose value is the quotient of the multiplicity of $(\Pi, \dim)$ in `invariant_counts` by $(\pi(b) - \pi(a))$, where π is the prime counting function.
 5. **Return** `dist`.
-

Remark 3.1.9. The output of Algorithm 2 enables us to eliminate some possible images immediately. For a given subgroup H , we check if each realized (characteristic polynomial, 1-eigenspace dimension pair) in the empirical local distribution is realized by an element of H . If not, then we may rule out H as a possibility.

Running Algorithm 2 on the example curve with $[a, b] = [10000, 20000]$, we compute the local distribution presented in the third column of Table 1.

3.1.2. Distribution Matching

By Lemma 1.3.5, there are exactly 99 distinct (characteristic polynomial, 1-eigenspace dimension) pairs. Thus, we embed the space of local distributions into $\mathbb{R}^{99}$ and use the L1 norm to measure similarity. Declaring this embedding an isometry defines a metric on the space of local distributions, allowing us to make Remark 3.1.8 precise.

Lemma 3.1.10. Fix any $a \in \mathbb{Z}$. There exists a constant N , depending on J , such that for all $b > N$, there is an inequality

$$|\mathcal{E}(J, a, b) - \mathcal{F}_{\text{im } \rho_{J,5}}| \leq |\mathcal{E}(J, a, b) - \mathcal{F}_G|$$

for any subgroup $G \leq \text{GSp}_4(\mathbb{F}_5)$ with surjective similitude character and order 2 element with similitude character -1 corresponding to complex conjugation, with equality only if $\mathcal{F}_{\text{im } \rho_{J,5}} = \mathcal{F}_G$.

Proof. Let $\mathbf{x}_b$ and $\mathbf{y}$ denote the images of the empirical distribution $\mathcal{E}(J, a, b)$ and the true distribution $\mathcal{F}_{\text{im } \rho_{J,5}}$, respectively, under the embedding into $\mathbb{R}^{99}$. Denote their coordinates $x_1, \dots, x_{99}$ and $y_1, \dots, y_{99}$, respectively. We first show that $\lim_{b \rightarrow \infty} \mathbf{x}_b = \mathbf{y}$.

The group $\text{GSp}_4(\mathbb{F}_5)$ has 156 conjugacy classes. Index the axes of $\mathbb{R}^{156}$ by conjugacy classes $\{\mathcal{C}\}$. Upgrading the notation of Theorem 1.3.3, let

$$P_{\mathcal{C}}(a, b) = \#\{\text{prime } p : \text{Frob}_p \in \mathcal{C}, p \in [a, b]\}.$$

Let $\mathbf{w}_b, \mathbf{z} \in \mathbb{R}^{156}$ have coordinates $w_{\mathcal{C}} = P_{\mathcal{C}}(a, b)/(\pi(b) - \pi(a))$ and $z_{\mathcal{C}} = \#\mathcal{C}/\#\text{GSp}_4(\mathbb{F}_5)$, respectively. The Chebotarev density theorem says that $w_{\mathcal{C}} \rightarrow z_{\mathcal{C}}$ as $b \rightarrow \infty$ for each $\mathcal{C}$. Thus, $\mathbf{w}_b \rightarrow \mathbf{z}$ as $b \rightarrow \infty$.

Let $p_{\mathcal{C}}(\lambda)$ denote the characteristic polynomial of the elements of $\mathcal{C}$, and let $d(\mathcal{C}, \lambda)$ denote the dimension of their λ -eigenspaces. There is a projection map $\mathbb{R}^{156} \rightarrow \mathbb{R}^{99}$ given by sending $e_{\mathcal{C}}$ to the basis vector corresponding to the pair $(p_{\mathcal{C}}, d(\mathcal{C}, 1))$. This projection map sends $\mathbf{w}_b \mapsto \mathbf{x}_b$ and $\mathbf{z} \mapsto \mathbf{y}$, so it follows from continuity of the projection that $\mathbf{x}_b \mapsto \mathbf{y}$ as $b \rightarrow \infty$.

Note that since $\text{GSp}_4(\mathbb{F}_5)$ is a finite group, its subgroup lattice is finite and there are finitely many distinct possibilities $\mathcal{F}_G$ which we must consider. This means that there is some minimal positive value

$$\delta = \min_G |\mathcal{F}_G - \mathcal{F}_{\text{im } \rho_{J,5}}|,$$

where G ranges over candidate subgroups. Since $\mathbf{x}_b \rightarrow \mathbf{y}$, i.e. $\mathcal{E}(J, a, b) \rightarrow \mathcal{F}_{\text{im } \rho_{J,5}}$, there exists a constant N such that $b > N$ implies the inequality $|\mathcal{E}(J, a, b) - \mathcal{F}_{\text{im } \rho_{J,5}}| < \frac{\delta}{2}$. The desired statement follows immediately. $\square$

Using this metric, we compute the list of possible images whose local distributions are closest to the empirical distribution. This is summarized in Algorithm 3.

Algorithm 3: An algorithm to compute likely images, given local distribution data.

Input: A set `possibilities` of possible images of Galois and a dictionary `frob.dist` encoding an empirical local distribution.

Output: The subset `candidates` of `possibilities` with minimal statistical distance to the empirical distribution and a parameter η indicating the estimated likelihood that `candidates` indeed contains $\text{im } \rho_{J,5}$.

1. **For** each subgroup in `possibilities` **do**

- (a) Compute the local distribution $\mathcal{F}_G$ associated to `subgroup`.
- (b) Compute the statistical distance between `frob.dist` and the subgroup's distribution.

2. Set **best_subgroups** to be the set of subgroups whose corresponding distributions achieve the minimum observed distance to the empirical distribution, and **second_subgroups** to be the set of subgroups whose distributions achieve the second lowest computed value.
3. Compute the log-likelihood difference (see Definition 3.1.13) $\eta = \mathcal{L}(J, a, b)$.
4. **Return candidates** and η .

Remark 3.1.11. By construction, Algorithm 3 will return a list of possible subgroups where all elements in the list have the same local distribution. Thus, by Remark 1.3.4, all subgroups in the list will have equal order, and by Lemma 3.1.4, the list will have length at most eight.

Lemma 3.1.12. Fix a Jacobian J and a lower bound a . There exists a constant $N \in \mathbb{N}$, depending on J , such that for every $b > N$ the following holds: if **candidates** is the output of running Algorithms 2 and 3 on J and $[a, b]$, then $\text{im } \rho_{J,5} \in \text{candidates}$.

Proof. Take N to be the value from the proof of Lemma 3.1.10. Since Algorithm 3 picks out the subgroups which minimize the statistical distance to the empirical local distribution, $\text{im } \rho_{J,5}$ will be an element of **candidates**. $\square$

In our example, applying Algorithm 3 returns subgroups 5.624.2 and 5.624.4. These both have the same local distribution. Putting this next to the empirical distribution obtained earlier, we get Table 1.

Characteristic Polynomial	1-eigenspace Dimension	Empirical Proportion	5.624.2 and 5.624.4 Proportion
$x^4 + 4$	1	12.88%	12.50%
$x^4 + 3x^2 + 1$	1	9.58%	9.17%
$x^4 + 3x^2 + 1$	2	2.32%	2.29%
$x^4 + x^3 + x^2 + x + 1$	1	4.16%	4.00%
$x^4 + x^3 + x^2 + x + 1$	2	0.97%	1.00%
$x^4 + x^3 + 3x^2 + 2x + 4$	0	6.58%	6.25%
$x^4 + x^3 + 3x^2 + 4x + 1$	1	6.49%	5.21%
$x^4 + x^3 + 4x^2 + 3x + 4$	0	4.36%	4.17%
$x^4 + 2x^3 + x^2 + 4x + 4$	0	3.58%	4.17%
$x^4 + 2x^3 + 3x^2 + x + 4$	0	4.07%	4.17%
$x^4 + 2x^3 + 3x^2 + 3x + 1$	1	2.52%	4.17%
$x^4 + 2x^3 + 4x^2 + 2x + 1$	1	3.58%	3.33%
$x^4 + 2x^3 + 4x^2 + 2x + 1$	2	0.97%	0.83%
$x^4 + 3x^3 + 2x^2 + 3x + 1$	1	4.94%	5.00%
$x^4 + 3x^3 + 2x^2 + 3x + 1$	2	0.68%	1.25%
$x^4 + 3x^3 + 2x^2 + 4x + 4$	0	5.91%	6.25%
$x^4 + 3x^3 + 3x^2 + 2x + 1$	1	4.65%	4.17%
$x^4 + 3x^3 + 4x^2 + x + 4$	0	4.45%	4.17%
$x^4 + 4x^3 + 4x + 1$	1	2.81%	3.33%
$x^4 + 4x^3 + 4x + 1$	2	0.97%	0.83%
$x^4 + 4x^3 + x^2 + 2x + 4$	0	3.58%	4.17%
$x^4 + 4x^3 + 2x^2 + 3x + 4$	0	4.94%	4.17%
$x^4 + 4x^3 + 3x^2 + x + 1$	1	5.03%	5.21%
$x^4 + x^3 + x^2 + x + 1$	3	0%	0.21%
$x^4 + x^3 + x^2 + x + 1$	4	0%	< 0.01%

Table 1: Comparing the empirical local distribution and the local distribution of the closest matches.

Thus at this point we assume that the set of possible images consists of $\{5.624.2, 5.624.4\}$. While the methods used for obtaining this local information are probabilistic and only guaranteed to be correct asymptotically, the data are empirically promising. If we assume that Frobenius elements are sampled

independently at random from a uniform distribution on elements of the true image, then the probability of obtaining the observed distribution from either 5.624.2 or 5.624.4 is $\exp(-3056.318)$. In contrast, the subgroups next most likely to yield the observed distribution are 5.312.1 and 5.312.2 (both have the same local distribution), and their corresponding probability is $\exp(-3461.531)$. The large discrepancy in likelihoods – a few hundred orders of magnitude – gives us confidence that the information derived in the procedure is accurate.

It is important to note that some distinct local distributions are very similar. For instance, the statistical distance between the local distributions of subgroups 5.74880.13 and 5.374400.24 is only $\frac{2}{125}$. Our algorithm requires a comparatively large number of sampled primes to distinguish such subgroups from one another. A desire to account for this motivates the condition in Step 3 of Algorithm 1 where we sample additional Frobenius elements until we reach a desired log-likelihood difference.

Definition 3.1.13. (log-likelihood ratio) Denote by $\{\mathcal{D}_i\}$ the set of local distributions corresponding to possible images of Galois. Let $\pi(n)$ denote the number of primes up to n . For J a Jacobian of a genus 2 curve and $[a, b]$ a range containing at least one prime, let $\mathcal{P}(\mathcal{D}_i, J, a, b)$ denote the probability, when sampling $\pi(b) - \pi(a)$ independent, identically distributed observations from $\mathcal{D}_i$, of obtaining a sample perfectly representative of the empirical local distribution $\mathcal{E}(J, a, b)$. Let m denote the value of the index i for which $\mathcal{P}(\mathcal{D}_i, J, a, b)$ is maximized. The *log-likelihood ratio* $\mathcal{L}(J, a, b)$ for J and $[a, b]$ may then be computed as

$$\mathcal{L}(J, a, b) = \log(\mathcal{P}(\mathcal{D}_m, J, a, b)) - \log\left(\max_{i \neq m} \mathcal{P}(\mathcal{D}_i, J, a, b)\right).$$

Using this notation, in our example Jacobian J we have

$$\mathcal{L}(J, 10000, 20000) = (-3056.318) - (-3461.531) = 405.213.$$

3.2. Global Information

3.2.1. Rational 5-torsion

By the fundamental theorem of Galois theory, $J(\mathbb{Q})[5]$ is exactly the subspace stabilized by $\text{im } \rho_{J,5}$. We can use this to constrain the image of Galois. We compute the rational 5-torsion order $n_1(J) := J(\mathbb{Q})[5]$ and filter for subgroups that stabilize exactly n_1 vectors in $\mathbb{F}_5^4$.

A first algorithm to compute the torsion part of the Mordell-Weil group is provided by Stoll [9, pp. 198–201], and there is an implementation in Magma [11]. The orbit sizes, and in particular the size of the stable set of a subgroup, is computed by Sutherland’s `GSpLattice` function.

In our running example, $J(\mathbb{Q}) \cong \mathbb{Z}$. In particular, $J(\mathbb{Q})[5]$ is trivial, so $\text{im } \rho_{J,5}$ cannot fix any non-zero vectors. Since neither 5.624.2 nor 5.624.4 fix a non-zero vector, this step does not narrow down the set of possibilities in our example. After checking for rational torsion, the set of possibilities remains $\{5.624.2, 5.624.4\}$.

3.2.2. Quadratic 5-torsion

A 5-torsion point is defined over a quadratic extension if and only if the point’s orbit under the action of Galois has size 2. Let $n_2(J)$ be the number of non-rational 5-torsion points defined over quadratic extensions, hereafter called *quadratic 5-torsion*. Then, $\text{im } \rho_{J,5}$ acts with exactly $n_2/2$ orbits of size 2. We thus perform a procedure analogous to the one outlined in Section 3.2.1. We compute $n_2(J)$, and filter for subgroups with $n_2/2$ orbits of size 2.

We determine n_2 in two phases. In the first phase we use information about the Jacobian mod various primes p to produce a short finite list of quadratic extensions of $\mathbb{Q}$ which could contain quadratic 5-torsion points. In the second phase we compute the torsion subgroup over each quadratic extension, and check how much 5-torsion is found.

Remark 3.2.1. By the Néron-Ogg-Shafarevich criterion, any torsion growth field has ramification only at primes of bad reduction and primes dividing the order of the torsion. In our specific setting, the following theorem gives a slightly stronger restriction.

Theorem 3.2.2 (Theorem 1.5 in [12]). Let A/K be a simple abelian variety over a number field K and let L be a quadratic extension of K . Assume that $A(L)_{\text{tors}} - A(K)_{\text{tors}}$ contains a point of prime order p . If for all primes $\mathfrak{p}$ of $\mathcal{O}_K$ with associated valuation $v_{\mathfrak{p}}$, we have that $v_{\mathfrak{p}}(p) < (p-1)/2$, then the primes of K that ramify in L are contained in

$$\{\mathfrak{p} \text{ is a prime of } \mathcal{O}_K : A/K \text{ has bad reduction at } \mathfrak{p}\}.$$

Corollary 3.2.3. Let C be a genus 2 curve with Jacobian $J = \text{Jac}(C)$ and conductor N . Let $L = \mathbb{Q}(\sqrt{d})$. Let $P \in J(L)$ be a simple quadratic torsion point. Then, we have $\text{rad}(\text{Disc}(L)) \mid N$.

Proof. Apply theorem 3.2.2 with $A = J$, $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{d})$, $p = 5$, and $\mathfrak{p} = (5)$. Each prime $q \mid \text{Disc}(L)$ ramifies in $\mathcal{O}_L$. Each of those primes are then primes of bad reduction by [12] Theorem 1.5, so they divide the conductor. $\square$

For any N , the set of quadratic fields L with bounded ramification is finite and easily enumerable. Thus, Corollary 3.2.3 allows us to compute a finite list of fields which could possibly contain quadratic 5-torsion. This list can be narrowed further using the following lemma.

Lemma 3.2.4. Let $K = \mathbb{Q}(\sqrt{d})$ be such that $\#J(K)[5] = 5^n$. Let $p \neq 5$ be a good rational prime which splits in K . Then, $J_{\mathbb{F}_p}$ admits 5^n rational 5-torsion points.

Proof. Let K be a quadratic field for which $J(K)[5] \neq 0$. Let $p \neq 5$ be a prime of good reduction with $p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$ for $\mathfrak{p}_1, \mathfrak{p}_2 \in \text{Spec}^*(\mathcal{O}_K)$. There is a reduction-mod- $\mathfrak{p}_1$ map $J(K) \rightarrow J(\mathbb{F}_{\mathfrak{p}_1}) \cong J(\mathbb{F}_p)$. Since the multiplication-by-5 map on $J/\mathbb{F}_{\mathfrak{p}_1}$ is étale, Hensel's lemma guarantees that reduction modulo $\mathfrak{p}_1$ is injective on 5-torsion. $\square$

The contrapositive of Lemma 3.2.4 allows us to sample primes and use them to preclude certain quadratic fields from containing simple quadratic 5-torsion. The first phase of the simple quadratic torsion determination is summarized in Algorithm 4.

Algorithm 4: An algorithm to compute where quadratic 5-torsion is possible.

Input: The Jacobian J of a hyperelliptic curve C with conductor N .

Output: A list of integers d , containing all d such that J may have 5-torsion over $\mathbb{Q}(\sqrt{d})$.

1. Initialize **possibilities** to the set of squarefree integers d such that $\text{rad}(\text{Disc}(\mathbb{Q}(\sqrt{d}))) \mid N$.
 2. **While** **possibilities** is not empty and for each good prime $p \in [10,000, 20,000]$ **do**
 - (a) Compute $J_{\mathbb{F}_p}(\mathbb{F}_p)$, the group of rational points of the Jacobian over $\mathbb{F}_p$. In practice, this data has already been computed in step 2 of Algorithm 2, so this step is computationally free.
 - (b) **If** $\#J_{\mathbb{F}_p}(\mathbb{F}_p)[5] = \#J(\mathbb{Q})[5]$ **then**
 - i. Remove from **possibilities** each d for which p splits in $\mathbb{Q}(\sqrt{d})$.
 3. **Return** **possibilities**.
-

If Algorithm 4 produces an empty list, we conclude there is no quadratic 5-torsion. If Algorithm 4 produces a non-empty list, then the second phase attempts to find quadratic 5-torsion in the remaining fields. For our example curve, running Algorithm 4 returns that there is one quadratic extension where there might be additional 5-torsion, $\mathbb{Q}(\sqrt{5})$. Thus we will check for additional 5-torsion over $\mathbb{Q}(\sqrt{5})$.

The Magma algorithm to compute the torsion subgroup of a genus 2 Jacobian is only implemented over $\mathbb{Q}$. We generalize the algorithm, given in [9], to number fields. We proceed by $\mathfrak{p}$ -adic lifting and rational recognition. This program may be of independent interest; it is available in the GitHub repository

https://GitHub.com/Aidan-Hennessey/genus2_jacobian_torsion.

Algorithm 5: An algorithm to determine whether an $\mathbb{F}_p$ -rational point $P \in J_{\mathbb{F}_p}(\mathbb{F}_p)$ of order m is the reduction of a torsion point $\tilde{P} \in J(K)[m]$.

Input: A model $y^2 = f(x)$ of a genus 2 curve C defined over a number field K , a prime $\mathfrak{p} \in \text{Spec}^*(\mathcal{O}_K)$, and a point $P \in J_{\mathbb{F}_p}(\mathbb{F}_p)$ of order greater than 2.

Output: An option – (**true**, $\tilde{P}$) if P is the reduction of a K -rational torsion point $\tilde{P}$, and **false** otherwise.

1. **If** $m = 1$ **then return true**.
 2. **If** $m = 2$ **then**
 - (a) The Mumford coordinates of P take the form $(u, 0)$ for u a quadratic factor of the reduction $f \bmod \mathfrak{p}$. Compute all K -rational quadratic factors $q_1, \dots, q_r$ of f .
 - (b) **If** $q_i \equiv u \bmod \mathfrak{p}$ for some i **then return true**. **Else return false**.
 3. Compute a bound h on the naïve height of torsion points $P \in J(K)$ (see Section 1.4).
 4. Compute the quantity r given in Theorem 2.0.6.
 5. Let P' denote the image of P on the Kummer surface. Compute a lift P_0 of P' to a point on the $\mathfrak{p}$ -adic Kummer surface, to precision r . This lift is in general no longer a torsion point.
 6. Let m denote the order of P . Compute a positive integer M which is both congruent to 1 mod m and congruent to 0 mod $\mathfrak{p}$.
 7. This next step only does anything for $m \geq 3$. This is why we handle $m = 1, 2$ separately at the start. Fix an affine patch of $\mathbb{P}^3$ containing P , so that we have a notion of scaling of points on the Kummer surface.
For $i \leq k = \lceil \log_2 r \rceil$ **do**
 Let $P_i = (MP_{i-1} - [M]P_{i-1})/(M - 1)$.
 In the above formula, MQ denotes the scaling of Q by a factor of M , and $[M]Q$ denotes the image of Q under the map on the Kummer surface induced by multiplication by M on the Jacobian.
 8. For the same reasons as in [9], the resulting point $P_k \in J(K_{\mathfrak{p}})$ is a torsion point of order m , known mod $\mathfrak{p}^r$. Let $[a_0 : a_1 : a_2 : a_3]$ be a choice of integral projective coordinates for P_k . Let $\{\alpha_1, \dots, \alpha_f\}$ be representatives in $\mathcal{O}_K$ for an $\mathbb{F}_p$ basis of the residue field $\mathbb{F}_p$, where p is the characteristic. Let $\{b_1, \dots, b_d\}$ be a $\mathbb{Z}$ -basis for $\mathfrak{p}^r$. Construct the lattice $L \subset (K \otimes \mathbb{R})^4$ generated by
 - (a) $\{(\alpha_i a_0, \alpha_i a_1, \alpha_i a_2, \alpha_i a_3) : i \in \{1, \dots, f\}\}$ and
 - (b) $\{(b_{i_1}, b_{i_2}, b_{i_3}, b_{i_4}) : i_1, i_2, i_3, i_4 \in \{1, \dots, d\}\}$.
 9. Perform LLL on L to obtain a short vector v . Pull v back to a 4-tuple (c_0, c_1, c_2, c_3) of elements of $\mathcal{O}_K$. By Theorem 2.0.6, if P is the reduction of a K -rational torsion point $\tilde{P}$, then $\tilde{P}$ has image $[c_0 : c_1 : c_2 : c_3]$ on the Kummer surface.
 10. **If** $[c_0 : c_1 : c_2 : c_3]$ is on the Kummer surface **then**
 - (a) Compute a preimage $\tilde{P} \in J(K)$.
 - (b) **If** $[m]\tilde{P} = 0_J$ **then return (true, $\tilde{P}$)**. **Else return false**.
- Else return false.**
-

Modulo the novelties of Algorithm 5, the following algorithm to compute the torsion subgroup proceeds nearly identically to that in [9].

Algorithm 6: An algorithm to compute the torsion subgroup of the Jacobian of a genus 2 curve over a number field.

Input: The Jacobian J of a genus 2 curve $C : y^2 = f(x)$ defined over a number field K

Output: The torsion subgroup $J(K)_{\text{Tors}}$

1. Reduce J mod several small good primes $\mathfrak{p}$, and compute $g = \gcd\{p^\infty \cdot \#J_{\mathbb{F}_p}(\mathbb{F}_p)\}$. Because reduction mod $\mathfrak{p}$ is injective on torsion away from p , this gcd will be a multiple of $\#J(K)_{\text{Tors}}$.
 2. **For** each prime $\ell \mid g$ **do**
 - (a) Find a good prime $p \neq \ell$ for which $\#J_{\mathbb{F}_p}(\mathbb{F}_p)[\ell^\infty]$ is minimal.
 - (b) Let $G = \#J_{\mathbb{F}_p}(\mathbb{F}_p)[\ell^\infty]$. Run Algorithm 7 on G with membership test Algorithm 5, yielding a subgroup H_ℓ of $J(K)$.
 3. **Return** $\bigoplus_\ell H_\ell$.
-

Algorithm 7: An algorithm to compute a subgroup of a known abelian group, given a membership test.

Input: An abelian group G , and a membership test T

Output: The subgroup $H \leq G$ consisting of those elements passing the membership test.

1. Let $B = \emptyset$.
 2. **While** $B \neq G \setminus \text{id}_G$ **do**
 - (a) Choose some non-identity $g \in G \setminus B$.
 - (b) **If** $T(g)$ **then**
 - i. Let $\overline{G} = G/\langle g \rangle$. Let $\overline{B} = \{\overline{b} : b \in B\}$, where overline denotes image in the quotient.
 - ii. Let $\overline{T}$ be a test that takes in an element $\overline{b} \in \overline{B}$, computes an arbitrary preimage b , and returns $T(b)$.
 - iii. Recursively run this algorithm on $\overline{G}$ and $\overline{T}$, except initialize B to $\overline{B}$, rather than $\emptyset$. Let $\overline{H}$ denote the output.
 - iv. **Return** the full preimage H of $\overline{H}$ under the quotient map $G \rightarrow G/\langle g \rangle$.
 - Else**

$$B \leftarrow B \cup \{h : g \in \langle h \rangle\}.$$
 3. **Return** $\{\text{id}_G\}$.
-

We now combine Algorithms 4 and 6 to compute $n_2(J)$.

Algorithm 8: An algorithm to count the quadratic 5-torsion of a Jacobian of a genus 2 curve over $\mathbb{Q}$.

Input: The Jacobian J of a genus 2 curve $C : y^2 = f(x)$ defined over $\mathbb{Q}$

Output: The number $n_2(J)$ of quadratic 5-torsion points of J

1. Run Algorithm 4 on input C , and initialize **possibilities** as the output. This is the list of quadratic extensions over which $J = \text{Jac}(C)$ possibly has quadratic 5-torsion.
2. Initialize $n_2 = 0$.
3. **For** each extension K in **possibilities** **do**
 - (a) Using Algorithm 6, compute $J(K)_{\text{Tors}}$, and in particular $J(K)[5]$.

(b) **If** $\#J(K)[5] > \#J(\mathbb{Q})[5]$ **then**
 $n_2 \leftarrow n_2 + (\#J(K)[5] - \#J(\mathbb{Q})[5]).$

4. **return** n_2 .

In our running example, we find that there is additional 5-torsion over $\mathbb{Q}(\sqrt{5})$. Specifically, Algorithm 8 yields that the point $(x^2 - 2x + 1, \sqrt{5}x - 2\sqrt{5})$ is a 5-torsion point over $\mathbb{Q}(\sqrt{5})$. This tells us that we can narrow down the list of possible images to subgroups compatible with quadratic 5-torsion. Checking each of the current candidates, $\{5.624.2, 5.624.4\}$, we find that only 5.624.2 is compatible with quadratic 5-torsion, so we conclude that the image is 5.624.2.

3.3. Proof of Correctness

Recall the Main Theorem.

Theorem (Main Theorem). Let $C/\mathbb{Q}$ be a genus 2 curve with Jacobian J . Fix any $a \in \mathbb{Z}$. There exists an effective constant N , depending only on C , such that there exists an algorithm sampling all primes in the range $[a, N]$ that produces a list containing the mod-5 image of Galois. The list is composed of at most eight subgroups of $\text{GSp}_4(\mathbb{F}_5)$, all of equal order.

Proof. We claim Algorithm 1, with $\nu = \infty$ suffices. Algorithm 1 proceeds by applying successive filters to a list of possible images. It suffices to verify that each filter retains the image $\text{im } \rho_{J,5}$, and that one filter returns only up to eight subgroups, all of the same order.

By Lemma 3.1.12, for adequately large N the output of Algorithm 3 contains the image of Galois. Furthermore, by Lemma 3.1.4, the output of Algorithm 3 is list of length at most eight. By the discussions at the start of Sections 3.2.1 and 3.2.2, the rational and quadratic torsion filters retain the image of Galois, provided the algorithms to compute the torsion subgroups are correct. Correctness of the torsion subgroup computation over $\mathbb{Q}$ is proven in [9]. The same proof largely goes through, modulo steps 8 and 9 of Algorithm 5. By Theorem 2.0.6, these steps are correct. □

4. Results

We ran Algorithm 1 on all genus 2 curves C in the [5] for which $\rho_{\text{Jac}(C),5}$ is likely not surjective, with prime range $[a, b] = [10^4, 10^5]$ and likelihood difference $\nu = 30$ (see Definition 3.1.13). Our choice of lower bound is chosen heuristically. One wants to exclude Frob_p from the computation for small p , because small p cannot be representative of the true local distribution. For example, for Frob_p to be the identity, $J_{\mathbb{F}_p}(\mathbb{F}_p)$ must have order dividing 625. Weil bounds therefore preclude small primes from representing the identity. Possible bias among small primes is one reason many effective Chebotarev-type statements consider the distribution of Frob_p for $p \in [N, 2N]$ as $N \rightarrow \infty$. A lower bound of 10000 is thus reasonable, because it is on the same order of magnitude as the size of our sample range. Our sample range size is largely determined by compute, but Lemma 4.0.1 indicates that this upper bound is sufficiently large.

This means we sampled Frob_p for all good primes $p \in [10000, 20000]$, a set of size up to 1033, and then, if needed to obtain our desired probability of error, sampled additional primes. To ensure the algorithm ran in a reasonable amount of time, we terminated sampling if we sampled all primes in $[10^4, 10^5]$, although this turned out to be unnecessary. In total, we ran Algorithm 1 on 3990 curves. On a server equipped with an Intel Core i9-12900K processor (16 cores, 24 threads) with a maximum clock speed of 5.2 GHz, the computation took 110 minutes for the typical curves and 345 minutes for the atypical curves. The majority of the algorithm's runtime is spent computing 1-eigenspace dimensions, i.e. computing $\#J_{\mathbb{F}_p}(\mathbb{F}_p)[5]$ for many p . When $x - 5$ is not a repeated factor of the characteristic polynomial of Frobenius, the characteristic polynomial determines the 1-eigenspace dimension. In the remaining cases, however, one needs to actually determine the Jordan block structure of Frob_p .

As previously mentioned, choosing a value of b large enough to attain the bounds required for an effective version of the Chebotarev Density Theorem would be prohibitively computationally expensive, hence our default choice of only sampling primes $p \in [10000, 20000]$. This choice is consistent with prior work, such as [3]. The code used for this computation can be found in the `LMFDB_imaging.magma` file in the GitHub repository. We now justify our choice of b, ν by estimating the likelihood of error.

Lemma 4.0.1. Let J be the Jacobian of a genus 2 curve listed in the LMFDB. Model the group elements $\{\text{Frob}_p\}$ as independent random variables drawn uniformly from the image of Galois. Let $\mathcal{A}$ be the geometric endomorphism algebra of J , and let $w_i = P(\text{im } \rho_{J,5} \in G(\mathcal{D}_i) \mid \mathcal{A})$ be the empirical prior probability of distribution $\mathcal{D}_i$ within category $\mathcal{A}$. Under this model, the probability that Algorithm 1 fails to return a set containing $\text{im } \rho_{J,5}$ is bounded above by $1.53 \cdot 10^{-10}$, or approximately 1 in 6.54 billion.

Proof. Algorithm 1 stops sampling Frobenius elements for a given curve when either (i) it achieves a log-likelihood difference of at least 30 or (ii) it samples all primes between 10,000 and 100,000. Crucially for the proof of this lemma, when we ran this algorithm, every single curve in fact terminated its sampling because it satisfied the log-likelihood condition. This condition implies that for the identified distribution $\mathcal{D}_1$ and any alternative $\mathcal{D}_i$ ($i \neq 1$), the likelihoods satisfy:

$$\mathcal{P}(\mathcal{D}_1, J, N) \geq e^{30} \mathcal{P}(\mathcal{D}_i, J, N) \quad \text{for all } i \neq 1.$$

Let w_i denote the empirical weight of distribution $\mathcal{D}_i$ within the endomorphism class of J . Based on our results across 3,990 curves, these weights vary from $w_{\max} = \frac{2188}{2468} \approx 0.88$ (for 5.650.1 in the $\mathbb{Q} \times \mathbb{Q}$ case) to a minimum of $w_{\min} = \frac{2}{2468} \approx 0.0008$ for rare images (like 5.0750.2 in the $\mathbb{Q} \times \mathbb{Q}$ case). Applying Bayes' rule with these non-uniform priors:

$$\begin{aligned} \mathbb{P}(\text{im } \rho_{J,5} \notin G(\mathcal{D}_1) \mid \text{data}) &= \frac{\sum_{i \neq 1} w_i \mathcal{P}(\mathcal{D}_i, J, N)}{w_1 \mathcal{P}(\mathcal{D}_1, J, N) + \sum_{i \neq 1} w_i \mathcal{P}(\mathcal{D}_i, J, N)} \\ &\leq \frac{\sum_{i \neq 1} w_i e^{-30} \mathcal{P}(\mathcal{D}_1, J, N)}{w_1 \mathcal{P}(\mathcal{D}_1, J, N) + \sum_{i \neq 1} w_i e^{-30} \mathcal{P}(\mathcal{D}_1, J, N)} \\ &= \frac{(1 - w_1) e^{-30}}{w_1 + (1 - w_1) e^{-30}}. \end{aligned}$$

The failure probability is a monotonically decreasing function of the prior w_1 . The “worst-case” scenario occurs when the algorithm correctly identifies a rare image with the minimum observed frequency $w_1 = 1/1234$. Substituting this minimum weight:

$$\mathbb{P}(\text{fail}) \leq \frac{(1 - \frac{1}{1234}) e^{-30}}{\frac{1}{1234} + (1 - \frac{1}{1234}) e^{-30}} \approx 3989 \cdot e^{-30} \approx 1.53 \cdot 10^{-10}.$$

Even when the true image is empirically rare, the high likelihood threshold $\nu = 30$ ensures that the posterior probability of error remains negligible. $\square$

Remark 4.0.2. Let $G(\mathcal{D}_i)$ denote the set of possible images with local distribution $\mathcal{D}_i$. More naively, one could set $w_i = \frac{\#G(\mathcal{D}_i)}{1125}$, recovering a uniform prior distribution. Running the same error probability computation for this distribution yields 1 in 9.5 billion.

Remark 4.0.3. As the conductor goes to infinity, the relative rate of appearance of between different subgroups blows up. Thus, any rigorously grounded prior distribution would have to take into account the conductor. With enough samples, the data collected should drown out any non-uniformity in the background distribution. So, any remotely reasonable prior distribution should make a decent enough starting point for error estimations. The similarity between the above two probabilities supports this.

For purposes of analysis, we split the data by (geometric) endomorphism algebra. Extra endomorphisms restrict the image of Galois, so sorting curves accordingly is a natural choice. Before detailing the results of our computations, we explain the significance of various common image labels.

4.1. Subgroups and labels

Sutherland's `GSpLattice` program computes the lattice of subgroups of $\text{GSp}_4(\mathbb{F}_5)$ with surjective similitude character, up to conjugacy. The program labels each subgroup in accordance with the LMFDB labeling system. Figure 1 gives a sublattice featuring the few most common images associated to the Jacobians having each class of endomorphism ring. We now describe each subgroup appearing in Figure 1.

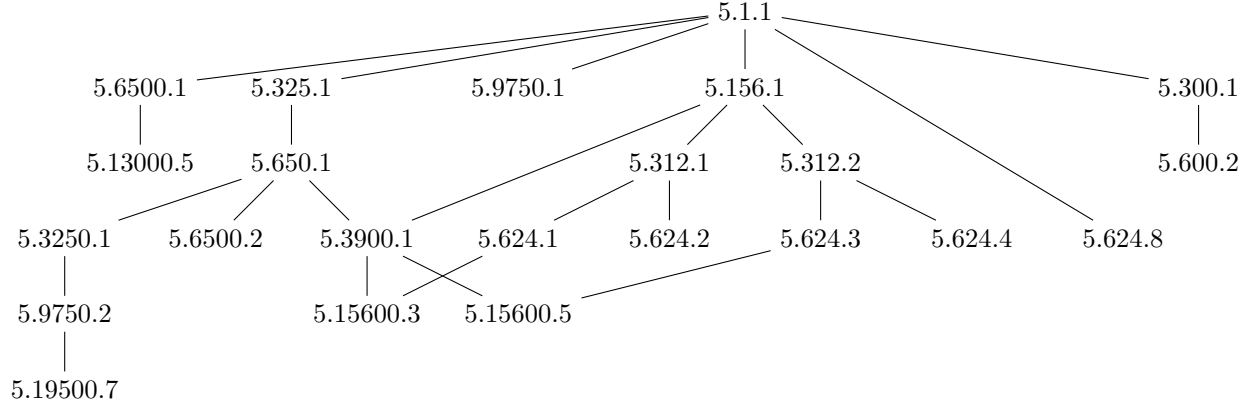


Figure 1: A lattice of common images of Galois

- (5.156.1) This is a maximal subgroup of $\mathrm{GSp}_4(\mathbb{F}_5)$, and specifically is the stabilizer of a 1-dimensional subspace. The group 5.156.1 also stabilizes a 3-dimensional subspace containing the 1-dimensional one. Thus, elements of 5.156.1 are block upper triangular with block sizes of (1, 2, 1). In other words, with the right choice of basis,

$$5.156.1 = \left\{ \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & 0 & * \end{bmatrix} \right\} \cap \mathrm{GSp}_4(\mathbb{F}_5).$$

A Jacobian admits a rational 5-isogeny if and only if its image is contained in 5.156.1.

- (5.312.1) This is the index 2 subgroup of 5.156.1 obtained by restricting the top-left entry to be ± 1 . A Jacobian admits torsion defined over a quadratic field if and only if $\mathrm{im} \rho_{J,5}$ is contained in 5.312.1.
- (5.624.1) This is the index 4 subgroup of 5.156.1 obtained by restricting the upper left entry to be 1. A Jacobian admits rational 5-torsion if and only if $\mathrm{im} \rho_{J,5} \leq 5.624.1$.
- (5.624.2) This is the index 2 subgroup of 5.312.1 characterized by $5.312.1 = 5.624.2 \times \{\pm 1\}$. The fact that 5.624.2 lacks the scalar matrix -1 implies that containment of an image in 5.624.2 may not be invariant under twisting - twists of Jacobians with image 5.624.2 may have image 5.312.1 [13, p. 7].
- (5.312.2) This subgroup is best understood by first understanding 5.624.3. This is the index 2 subgroup of 5.156.2 obtained by restricting the bottom-right entry to be ± 1 . There is a quotient q onto the bottom right entry. If 5.312.2 is the image attached to J , then $q \circ \rho$, by virtue of being a continuous homomorphism $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \{\pm 1\}$, is a quadratic character $\chi_{\mathbb{Q}(\sqrt{d})}$. One may thus consider the restricted Galois representation $\rho_{J,5} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{d})) \rightarrow \mathrm{GSp}_4(\mathbb{F}_5)$. The image of this restricted Galois representation is contained in 5.624.3. For the same reasons as discussed in the description of 5.624.3, it follows that such a Jacobian lacks quadratic 5-torsion but is isogenous to a variety with quadratic 5-torsion. 5.312.2 is the maximal possible image subject to this property.
- (5.624.3) This is the index 4 subgroup of 5.156.1 obtained by restricting the bottom right entry to be 1. Every element of this group has a non-trivial 1-eigenspace. Thus, Jacobians with image 5.624.3 have 5-torsion mod every prime of good reduction, but not over $\mathbb{Q}$. It follows by a theorem of Katz [14, p. 483] that any such Jacobian is isogenous to a variety with rational 5-torsion. Conversely, any Jacobian with that property has image contained in 5.624.3.
- (5.624.4) This is the index 2 subgroup of 5.312.2 characterized by $5.312.2 = 5.624.4 \times \{\pm 1\}$. Like 5.624.2, this subgroup may contain the image attached to a Jacobian but not to all twists thereof.

- (5.325.1) This subgroup is maximal, and is the normalizer of 5.650.1. In a symplectic basis, 5.325.1 is

$$5.325.1 = \left\{ \begin{bmatrix} * & * & 0 & 0 \\ * & * & 0 & 0 \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{bmatrix}, \begin{bmatrix} 0 & 0 & * & * \\ 0 & 0 & * & * \\ * & * & 0 & 0 \\ * & * & 0 & 0 \end{bmatrix} \right\} \cap \mathrm{GSp}_4(\mathbb{F}_5).$$

Jacobians admitting a quadratically-defined isogeny of degree prime to 5 to a product of non-isogenous elliptic curves have image contained in 5.325.1.

- (5.650.1) This is the group of elements preserving each summand in a direct sum decomposition of $J(\mathbb{Q})[5]$ into two nondegenerate 2-dimensional subspaces each defined individually over $\mathbb{F}_5$. If there is a rationally-defined isogeny $J \rightarrow E_1 \times E_2$ of degree relatively prime to 5 from a Jacobian J to a product $E_1 \times E_2$ of non-isogenous elliptic curves, then $\mathrm{im} \rho_{J,5} \leq 5.650.1$. The converse is notably false — see Section 4.2.
- (5.3250.1) This is the maximal subgroup of 5.650.1 whose action on one of the preserved 2-dimensional subspaces is the same as that of the exceptional maximal subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$ with label 5.5.1.
- (5.9750.2) This is the subgroup of 5.650.1 whose action on one of the preserved 2-dimensional subspaces is the same as that of the normalizer of a split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$. 5.9750.2 is not maximal in 5.650.1; it is index 3 in the intermediary subgroup 5.3250.1. This corresponds to the fact that the normalizer of a split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$ is not maximal, but instead contained in an exceptional maximal subgroup.
- (5.19500.7) This is an index 2 subgroup of 5.9750.2. Specifically, this is the subgroup of 5.650.1 whose action on one of the preserved 2-dimensional subspaces is the same as that of G , the intersection of the normalizer of a split Cartan and the normalizer of a non-split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$. Concretely, G is the index 2 subgroup of the normalizer of a split Cartan for which the diagonal elements all have square determinant and the off-diagonal elements all have non-square determinant.
- (5.6500.2) This is the subgroup of 5.650.1 whose action on one of the preserved 2-dimensional subspaces is the same as that of the normalizer of a non-split Cartan subgroup of $\mathrm{GL}_2(\mathbb{F}_5)$.
- (5.3900.1) This subgroup is the intersection of 5.156.1 and 5.650.1. With the right choice of basis, one may express 5.3900.1 as

$$5.3900.1 = \left\{ \begin{bmatrix} * & * & 0 & 0 \\ 0 & * & 0 & 0 \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{bmatrix} \right\} \cap \mathrm{GSp}_4(\mathbb{F}_5).$$

- (5.15600.3) This subgroup is the intersection of 5.624.1 and 5.650.1. It is the index-4 subgroup of 5.3900.1 for which the upper left entry is 1.
- (5.15600.5) This subgroup is the intersection of 5.624.3 and 5.650.1. It is the index-4 subgroup of 5.3900.1 for which the entry in the second row and second column is 1.
- (5.624.8) This is an index 4 subgroup of the maximal subgroup 5.156.2, which is the stabilizer of a 2-dimensional isotropic subspace. With the right choice of basis, 5.624.8 is the subgroup consisting of all block upper triangular elements with blocks of size 2 such that the two diagonal blocks are equal. Alternatively, 5.624.8 is the intersection $Z_{\mathrm{GL}_4(\mathbb{F}_5)}(\varphi) \cap \mathrm{GSp}_4(\mathbb{F}_5)$ of $\mathrm{GSp}_4(\mathbb{F}_5)$ with the centralizer in $\mathrm{GL}_4(\mathbb{F}_5)$ of an element φ with the following Jordan canonical form:

$$\varphi \simeq \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

The element $\varphi \in \mathrm{GL}_4(\mathbb{F}_5)$ is so named because it shares a minimal polynomial with the golden ratio.

- (5.300.1) This is a maximal subgroup. This is the normalizer of 5.600.2.
- (5.600.2) This is an index 2 subgroup of 5.300.1. It is the subgroup preserving each summand in a direct sum decomposition of $\mathbb{F}_5^4$ into two nondegenerate 2-dimensional subspaces defined jointly, but not individually, over $\mathbb{F}_5$. Alternatively, 5.600.2 is the intersection $Z_{\mathrm{GL}_4(\mathbb{F}_5)}(M) \cap \mathrm{GSp}_4(\mathbb{F}_5)$ of $\mathrm{GSp}_4(\mathbb{F}_5)$ with the centralizer in $\mathrm{GL}_4(\mathbb{F}_5)$ of an element M with the following Jordan canonical form:

$$M \simeq \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 4 \end{bmatrix}.$$

- (5.9750.1) This is a maximal subgroup of $\mathrm{GSp}_4(\mathbb{F}_5)$. It is the normalizer of the subgroup preserving a direct sum decomposition of $\mathrm{GSp}_4(\mathbb{F}_5)$ into two 2-dimensional isotropic subspaces, each defined individually over $\mathbb{F}_5$. This group also coincides with the exceptional maximal subgroup G_{1920} described in [3, p. 7].
- (5.6500.1) This is a maximal subgroup of $\mathrm{GSp}_4(\mathbb{F}_5)$. It is the normalizer of 5.13000.5.
- (5.13000.5) This has index 2 in the maximal subgroup 5.6500.1. It is the subgroup preserving a direct sum decomposition of $\mathrm{GSp}_4(\mathbb{F}_5)$ into two 2-dimensional isotropic subspaces defined jointly, but not individually, over $\mathbb{F}_5$.

4.2. Endomorphism algebra $\mathbb{Q}$

Of the 3990 Jacobians with non-surjective mod-5 image of Galois, 939 have endomorphism ring $\mathbb{Z}$. Among these, we determined a precise likely image for 935 curves.

The remaining four curves are as follows.

- The curve given by $y^2 = 4x^5 + 5x^4 - 10x^3 - 25x^2 + 30x - 3$ (37500.a.37500.1) is undetermined between subgroups 5.14976.1 and 5.14976.3.
- The curve given by $y^2 = 4x^5 - 4x^4 - 16x^3 - 79x^2 - 76x - 20$ (240250.a.240250.1) is undetermined between subgroups 5.14976.10 and 5.14976.11.
- The curve given by $y^2 = -4x^6 - 12x^5 - 15x^4 + 10x^2 + 4x - 3$ (400000.a.400000.1) is undetermined between subgroups 5.12480.19 and 5.12480.17. Unlike for the other three curves in this list, the two remaining candidates for this curve's image are abstractly isomorphic.
- The curve given by $y^2 = 4x^5 + 5x^4 - 10x^3 - 5x^2 + 10x + 5$ (787500.a.787500.1) is undetermined between 5.14976.5 and 5.14976.7.

By far, the most common non-surjective image of Galois for Jacobians with endomorphism ring $\mathbb{Z}$ is 5.624.1. Thus, the most common obstruction to surjectivity of the mod-5 image of Galois is presence of rational 5-torsion. The next most common obstruction is presence of quadratic 5-torsion. After these, the most common obstructions are the existence of isogenous varieties with the just-described obstructions. Perhaps surprisingly, it appears to be quite rare for Jacobians to admit 5-isogenies without an additional obstruction; only 16 Jacobians were computed to have mod-5 image of Galois 5.156.1. We expect that this behavior is a phenomenon only in small conductor, and that image 5.156.1 would be more common asymptotically.

Most Jacobians with image 5.650.1 are isogenous to a product of elliptic curves. The sole exception among all 3990 computed images is the curve $C : y^2 = x^6 + 8x^5 + 20x^4 + 20x^3 - 8x - 4$ (label 600000.b.600000.1 in the LMFDB). The Jacobian J has geometric endomorphism ring $\mathbb{Z}$, and thus is geometrically simple. Despite this, J has mod-5 image of Galois 5.650.1.

A summary of the results is available in Figure 2, with the full data in Table 2 in Appendix A.

Images of Typical Curves

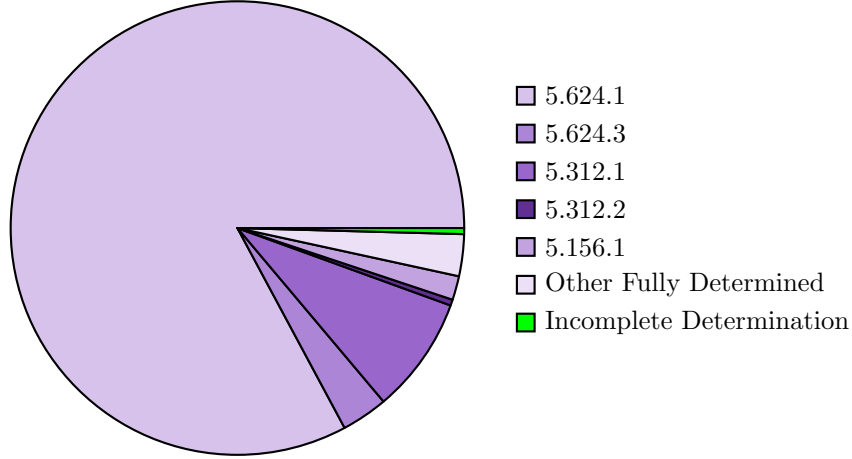


Figure 2: A breakdown of the results of Algorithm 1 on typical curves.

4.3. Endomorphism algebra $\mathbb{Q} \times \mathbb{Q}$

A majority (2468, to be precise) of the computed images were of Jacobians with endomorphism algebra $\mathbb{Q} \times \mathbb{Q}$. Such Jacobians are isogenous to products of non-isogenous elliptic curves lacking complex multiplication. We were able to compute the precise mod-5 image of Galois for all but ten of the Jacobians in this category. All such Jacobians are undecided between 5.14976.2 and 5.14976.4.

By far, the most common image in this category is 5.650.1, attached to just over 88% of Jacobians with geometric endomorphism algebra $\mathbb{Q} \times \mathbb{Q}$. The next four most common images, in order, are 5.325.1, 5.15600.3, 15600.5, and 5.3900.1. These five images account for over 98% of Jacobians in this category.

As an example, consider the curve $C : y^2 = x^6 - 4x^4 - 12x^3 - 16x^2 - 12x - 4$ (label [1573.a.1573.1](#) in the LMFDB) with Jacobian J . We compute that $\text{im } \rho_{J,5} = 5.15600.3$. From our description of this subgroup, we may expect a rational n -isogeny, for n prime to 5, from J to a product $E_1 \times E_2$ of non-isogenous elliptic curves for which $\rho_{E_1,5}$ is surjective and $\rho_{E_2,5}$ is the stabilizer of a vector $v \in \mathbb{F}_5^2$.

We indeed see exactly this; $J \simeq E_1 \times E_2$, for elliptic curves E_1 and E_2 given by $E_1 : y^2 + y = x^3 - x^2 - x - 2$ and $E_2 : y^2 + y = x^3 - x^2$ (Cremona labels [143.a.1](#) and [11.a.3](#) respectively). An LMFDB lookup confirms that $\rho_{E_1,5}$ and $\rho_{E_2,5}$ are as we predict from our computations.

Most Jacobians with image 5.624.8 have a geometric endomorphism algebra isomorphic to a real quadratic field, but there are also seven Jacobians in the $\mathbb{Q} \times \mathbb{Q}$ category with image 5.624.8. The example with smallest conductor is the Jacobian of the curve $y^2 = 4x^6 + 4x^5 + 17x^4 + 26x^3 + 25x^2 + 36x + 20$ ([2028.a.64896.1](#)). These curves admit isogenies of degree a multiple of 5 to products of elliptic curves.

4.4. Endomorphism algebra $\mathbb{Q} \times CM$

The third most common category of Jacobians is that consisting of Jacobians whose geometric endomorphism algebra is $\mathbb{Q} \times K$ for K an imaginary quadratic field. Jacobians in this category are isogenous to a product of elliptic curves, exactly one of which admits complex multiplication. Our algorithm computes unique images for every Jacobian in this category.

The vast majority of these Jacobians have image 5.9750.2 or image 5.6500.2. Of the remaining 8 Jacobians, 5 have image 5.19500.7.

The failure of Jacobians in this category to have surjective image of Galois is often a well-understood consequence of how they split. For example, the curve in this category of minimal conductor is $C : y^2 = x^6 - 6x^4 + x^2 + 28$ (LMFDB label [448.a.448.2](#)). The Jacobian J of C splits via a rational isogeny into a product $E_1 \times E_2$ of elliptic curves given by $E_1 : y^2 + xy + y = x^3 - x$ and $E_2 : y^2 = x^3 - x$, according to the [\[5\]](#). Since C is bielliptic, the isogeny $J \rightarrow E_1 \times E_2$ is of degree prime to 5. The first curve is not CM, and has maximal image of Galois $\rho_{E_1,5} \cong \text{GL}_2(\mathbb{F}_5)$. The second curve, E_2 , has potential complex multiplication; its geometric endomorphism ring is isomorphic to $\mathbb{Z}[i]$. The extra endomorphism i , given by

$(x, y) \mapsto (-x, iy)$, is defined over $\mathbb{Q}(i)$. The endomorphism i acts on $E_2[5]$ as $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, so there is containment $\rho_{E_2,5}(\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(i))) \leq Z_{\text{GL}_2(\mathbb{F}_5)}(\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix})$ of the image of the restricted Galois representation in the centralizer of i . As a linear transformation of $E_2[5]$, the extra endomorphism i has centralizer H a split Cartan subgroup of $\text{GL}_2(\mathbb{F}_5)$. The extension $\mathbb{Q}(i)/\mathbb{Q}$ is of course Galois, so H is a normal subgroup of the image of Galois. Equivalently, the image of Galois is contained in the normalizer $N(H)$. Since the isogeny $J \rightarrow E_1 \times E_2$ has degree prime to 5,

$$\text{im } \rho_{J,5} \leq (\text{im } \rho_{E_1,5} \times \text{im } \rho_{E_2,5}) \cap \text{GSp}_4(\mathbb{F}_5) \leq (\text{GL}_2(\mathbb{F}_5) \times N(H)) \cap \text{GSp}_4(\mathbb{F}_5) = 5.9750.2.$$

Our computation of $\text{im } \rho_{J,5}$ yields the group 5.9750.2, which coincides with the upper bound established by the geometric obstructions of E_1 and E_2 . This result indicates that the image of Galois is as large as possible given the splitting of J and the CM of E_2 , confirming that there are no further obstructions to surjectivity.

4.5. Real quadratic endomorphism algebra

There are 116 computed Jacobians with geometric endomorphism algebras isomorphic to real quadratic fields. Real quadratic fields lack 0-divisors, so Jacobians in this category are simple.

Among these, the most common image is 5.624.8. That is plausible, for the following reason. Many Jacobians in the LMFDB with real multiplication happen to have endomorphism ring $\mathbb{Z}\left[\frac{\sqrt{5}+1}{2}\right]$. This may be a product of a phenomenon where, in small conductor, the endomorphism ring is of small discriminant. In any case, when $\text{End}(J) \cong \mathbb{Z}\left[\frac{\sqrt{5}+1}{2}\right]$, the extra endomorphism generating $\mathbb{Z}\left[\frac{\sqrt{5}+1}{2}\right]$, which we denote φ , has minimal polynomial $x^2 - x - 1 \equiv (x - 3)^2 \pmod{5}$. Thus, φ must have only 3 as an eigenvalue, and its Jordan canonical form must be comprised of Jordan blocks of sizes 2, 1, and 1, or of sizes 2 and 2. When φ is defined over $\mathbb{Q}$, the image of Galois must be contained in the centralizer of φ . In the case where φ consists of two Jordan blocks of size 2, the centralizer of φ is exactly the subgroup 5.624.8.

The next most common images are 5.600.2 and 5.300.1. These are plausible for similar reasons. For $d \equiv 2, 3 \pmod{5}$, any order of $\mathbb{Q}(\sqrt{d})$ is generated by an element with minimal polynomial congruent to $x^2 + x + 2 \pmod{5}$. Any element of $\text{GL}_4(\mathbb{F}_5)$ with minimal polynomial $x^2 + x + 2$ is conjugate to the matrix M given in the description of 5.300.1. Suppose the rationally-defined endomorphism algebra of J is $\mathbb{Q}(\sqrt{d})$ for d non-square mod 5. In this case, there is an endomorphism which acts as M on the 5-torsion. The subgroup 5.600.2 is the centralizer of M , so the image of Galois is contained in 5.600.2. If the extra endomorphism is defined over a quadratic extension, then we similarly obtain that the image of Galois is contained in 5.300.1.

Our algorithm has a slightly lower success rate with Jacobians of this type compared to the most common types; we are unable to identify a precise image for 9 of the 116 Jacobians in this category.

4.6. Endomorphism algebra $M_{2 \times 2}(\mathbb{Q})$

Jacobians with (geometric) endomorphism algebra $M_{2 \times 2}(\mathbb{Q})$ are isogenous to a square $E \times E$ of an elliptic curve lacking complex multiplication. This category of Jacobians gives our algorithm the most trouble; our algorithm computes a precise likely image for only 108 of the 147 Jacobians in this category.

The most common output of our algorithm for this category is 5.13000.5. The next most common output is $\{5.39000.2, 5.39000.7, 5.39000.8\}$, which accounts for 20 of the 39 curves in this category for which the precise likely image is unknown. This is unfortunate in that the three candidates feature large differences; for example they are contained in different maximal subgroups. In particular, they are not locally conjugate. Below is the full list of containments.

- 5.39000.2 is contained in the maximal subgroups 5.156.2, 5.300.1, 5.325.1, and 5.9750.1. Note that 5.156.2 is not included in Figure 1; see the description of 5.624.8 for a description of 5.156.2.
- 5.39000.7 is contained in the maximal subgroups 5.300.1, 5.325.1, and 5.6500.1. 5.39000.7 is the intersection of (suitably chosen representatives of) 5.650.1 and 5.6500.1.
- 5.39000.8 is contained in the maximal subgroups 5.325.1 and 5.9750.1. 5.39000.8 is the intersection of (suitably chosen representatives of) 5.650.1 and 5.9750.1.

4.7. Other endomorphism algebras

In the LMFDB, 17 genus 2 curves have Jacobians with geometric endomorphism algebras which are not of any of the so-far covered forms. Our algorithm determined a precise likely image of Galois for the Jacobians of 8 of these curves. Our algorithm does not employ knowledge of the endomorphism ring of the Jacobian. Because these Jacobians have comparatively large endomorphism rings, incorporating such information would likely yield an outsized improvement for curves in this category and for curves with endomorphism algebra $M_{2 \times 2}(\mathbb{Q})$.

Appendix A: Computational Results

Below are the results of our computations, broken up by endomorphism algebra type. Minimal-conductor examples are given as LMFDB labels.

Table 2: Image counts for mod-5 Galois images of LMFDB curves with typical endomorphism ring

Likely image(s)	Count	Minimal-conductor example
$\{5.624.1\}$	777	277.a.277.1
$\{5.312.1\}$	78	4293.a.4293.1
$\{5.624.3\}$	31	523.a.523.2
$\{5.624.4\}$	19	6625.c.33125.1
$\{5.156.1\}$	16	8960.c.17920.1
$\{5.312.2\}$	4	4672.a.9344.1
$\{5.624.2\}$	4	431250.a.431250.1
$\{5.9360.15\}$	2	100000.a.200000.1
$\{5.78000.22\}$	1	9125.a.228125.1
$\{5.3744.1\}$	1	18928.a.984256.1
$\{5.14976.1, 5.14976.3\}$	1	37500.a.37500.1
$\{5.6240.4\}$	1	84375.a.84375.1
$\{5.14976.10, 5.14976.11\}$	1	240250.a.240250.1
$\{5.12480.19, 5.12480.17\}$	1	400000.a.400000.1
$\{5.650.1\}$	1	600000.b.600000.1
$\{5.14976.5, 5.14976.7\}$	1	787500.a.787500.1

Table 3: Image counts for mod-5 Galois images of LMFDB curves with endomorphism algebra $\mathbb{Q} \times \mathbb{Q}$

Likely image(s)	Count	Minimal-conductor example
$\{5.650.1\}$	2188	294.a.294.1
$\{5.325.1\}$	125	1088.a.1088.1
$\{5.15600.3\}$	68	363.a.11979.1
$\{5.15600.5\}$	30	847.d.847.1
$\{5.3900.1\}$	19	8788.a.17576.1
$\{5.14976.2, 5.14976.4\}$	10	726.a.1452.1
$\{5.15600.6\}$	8	1125.a.151875.1
$\{5.624.8\}$	7	2028.a.64896.1
$\{5.78000.22\}$	4	363.a.43923.1
$\{5.3250.1\}$	3	52488.a.629856.1
$\{5.234000.3\}$	2	40000.a.160000.1
$\{5.234000.4\}$	2	40000.c.200000.1
$\{5.9750.2\}$	2	520000.a.520000.1

Table 4: Image counts for mod-5 Galois images of LMFDB curves with endomorphism algebra $\mathbb{Q} \times K$ for K an imaginary quadratic field.

Likely image(s)	Count	Minimal-conductor example
$\{5.9750.2\}$	163	448.a.448.2
$\{5.6500.2\}$	132	686.a.686.1
$\{5.19500.7\}$	5	67500.a.810000.1
$\{5.234000.3\}$	1	1331.a.1331.1
$\{5.468000.4\}$	1	16875.a.84375.1
$\{5.468000.1\}$	1	16875.b.151875.1

Table 5: Image Counts for mod-5 Galois Images of LMFDB Curves with real quadratic endomorphism algebras

Likely image(s)	Count	Minimal-conductor example
$\{5.624.8\}$	55	529.a.529.1
$\{5.600.2\}$	29	841.a.841.1
$\{5.300.1\}$	10	36864.a.36864.1
$\{5.74880.4\}$	4	961.a.961.2
$\{5.14976.2, 5.14976.4\}$	4	7569.a.68121.1
$\{5.624.7\}$	4	180625.a.903125.1
$\{5.14976.6, 5.14976.8\}$	2	62500.a.1000000.1
$\{5.74880.16\}$	1	961.a.961.1
$\{5.74880.5\}$	1	961.a.923521.1
$\{5.74880.13\}$	1	15625.a.15625.1
$\{5.374400.2, 5.374400.10\}$	1	12500.a.12500.1
$\{5.374400.4\}$	1	12500.b.50000.1
$\{5.374400.5\}$	1	50000.b.800000.1
$\{5.14976.1, 5.14976.3\}$	1	112500.a.450000.1
$\{5.74880.21, 5.74880.14\}$	1	378125.a.378125.1

Table 6: Image counts for mod-5 Galois images of LMFDB curves with endomorphism algebra $M_{2 \times 2}(\mathbb{Q})$

Likely image(s)	Count	Minimal-conductor example
$\{5.13000.5\}$	51	169.a.169.1
$\{5.39000.8, 5.39000.7, 5.39000.2\}$	20	69696.c.627264.1
$\{5.6500.1\}$	13	15552.c.746496.1
$\{5.9750.1\}$	12	4608.c.27648.1
$\{5.78000.4, 5.78000.24\}$	8	196.a.21952.1
$\{5.26000.1\}$	7	324.a.648.1
$\{5.19500.3, 5.19500.2\}$	6	102400.b.102400.1
$\{5.19500.1\}$	4	6400.f.64000.1
$\{5.39000.1\}$	4	25600.d.128000.1
$\{5.13000.4\}$	4	6075.a.18225.1
$\{5.39000.17, 5.39000.6, 5.39000.5\}$	2	3969.d.250047.1
$\{5.374400.4\}$	2	2500.a.50000.1
$\{5.234000.24, 5.234000.23, 5.234000.18\}$	2	589824.a.589824.1
$\{5.13000.8\}$	2	2187.a.6561.1
$\{5.58500.2\}$	2	262144.b.524288.1
$\{5.468000.25\}$	1	256.a.512.1
$\{5.936000.12, 5.936000.9, 5.936000.13\}$	1	576.a.576.1
$\{5.234000.34\}$	1	4096.e.524288.1
$\{5.19500.4\}$	1	8192.b.131072.1

$\{5.48750.1\}$	1	12544.g.175616.1
$\{5.19500.8\}$	1	12800.c.128000.1
$\{5.65000.5\}$	1	26244.d.314928.1
$\{5.13000.3\}$	1	26244.e.472392.1

Table 7: Image counts for mod-5 Galois images of LMFDB curves with endomorphism algebra $M_{2 \times 2}(K)$ for K an imaginary quadratic field

Likely image(s)	Count	Minimal-conductor example
$\{5.390000.12, 5.390000.18, 5.390000.14, 5.390000.13, 5.390000.7\}$	4	4096.b.65536.1
$\{5.585000.14, 5.585000.22\}$	1	5184.a.46656.1
$\{5.1560000.7\}$	1	40000.e.200000.1
$\{5.260000.11, 5.260000.16\}$	1	2916.b.11664.1
$\{5.130000.16\}$	1	11664.a.11664.1

Table 8: Image counts for mod-5 Galois images of LMFDB curves with endomorphism algebra a quartic CM field

Likely image(s)	Count	Minimal-conductor example
$\{5.374400.5\}$	2	10000.b.800000.1
$\{5.1872000.10, 5.1872000.1\}$	1	3125.a.3125.1
$\{5.90000.1\}$	1	28561.a.371293.1
$\{5.187200.52\}$	1	50625.a.253125.1
$\{5.187200.52\}$	1	160000.c.800000.1

Table 9: Image counts for mod-5 Galois images of LMFDB Curves with non-split quaternion algebras as endomorphism algebras

Likely image(s)	Count	Minimal-conductor example
$\{5.19500.3, 5.19500.2\}$	2	262144.d.524288.1
$\{5.9750.1\}$	1	20736.l.373248.1

In addition to all the forms described above, it is possible for a Jacobian of a genus 2 curve to have endomorphism algebra $K \times K'$ for K and K' distinct imaginary quadratic fields. However, at the time of writing, there are no such curves in the LMFDB.

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